Government of Karnataka

Dept. of Collegiate &Technical Education

Government Engineering College, Raichur-584135

(Affiliated to Visvesvaraya Technological University)

Department of Electronics and Communication Engineering

Analog and Digital Systems Design Laboratory Manual

for

III sem BE (E&C)

Lab Incharge: MADHU YB

VISION:
Impart quality education to create world class technocrats and entrepreneurs with self-reliance to impart new ideas and innovations to excel in technical educationand scientific research in Electronics & Communication Engineering.

MISSION:
- M1: To Develop and deliver Quality academic programs in Emerging and innovative field of Engineering and science to empower the students to meet highest standards.
- M2: To provide domain affirmation, activities to add moral values and social awareness for the improvement in the effective engineering development, implementation and management of exploration based learning and to develop the thinking capability of young minds.
- M3: To create Centre of Excellence by establishing the skill development and incubation centers to meet global research challenges.

PROGRAM OUTCOME (PO):

PO-1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO-2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO-3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO-4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO-5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO-6: The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO-7: Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of need for sustainable development.
PO-8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO-9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO-10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO-11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO-12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

PROGRAM SPECIFIC OUTCOME (PSO):

- PSO1: Specify, design, build and test analog and digital systems for signal processing including multimedia applications, using suitable components or simulation tools.
- PSO2: Understand and architect wired and wireless analog and digital communication systems as per specifications, and determine their performance.

PROGRAM EDUCATIONAL OBJECTIVES (PEO):

- PEO-1: Excel in professional education in electronics & communication engineering with quality education.
- PEO-2: Exhibit professionalism, Ethics, communication skills, team work, self dependence and right attitude in the profession as well adapt to the changes by continuous learning.
- PEO-3: Bring out the solution to real time problems in electronics & communication engineering developments.

Guidelines for Teachers
To implement an outcome-based education (OBE), the level of knowledgeand skills of the students should be enhanced. Teachers are responsible forthe proper implementation of OBE. Some of the responsibilities (not limitedto) for the teachers in an OBE system are as follows:
- Within reasonable constraints, teachers should manage time to the advantage of all students.
- Teachers should assess students on certain defined criteria without considering any other potential ineligibility to discriminate them.
- Teachers should foster the learning abilities of the students.
- Teachers should ensure that all the students are equipped with knowledge and competence when they finish their education.

- Teachers should encourage students to improve their peak performancecapabilities.
- Teachers should facilitate and encourage group work and team work.
- Teachers should follow Blooms taxonomy in every assessment.
- Students' learning should be connected with practical and real life situations.

BLOOMS TAXONOMY

LEVEL		Teacher should check students' ability to	Student should be able to	Possible mode of assessment
CREATE	L6	Create	Design or Create	Mini Project
EVALUATE	L5	Justify	Argue or Defend	Assignment
ANALYSE	L4	Distinguish	Differentiate or Distinguish	Project/ Lab Methodology
APPLY	L3	Use Information	Operate or Demonstrate	Technical Presentation
UNDERSTAND	L2	Explain Ideas	Explain or Classify	Presentation or Seminar
REMEMBER	L1	Recall	Define or Recall	Quiz

Guidelines for Students:

Students should take equal responsibility in implementing an OBE. Some ofthe responsibilities for the students in an OBE system are as follows:

- Students should be aware of each CO before the start of a unit.
- Students should be well aware of each CO before the start of the course.
- Students should be well aware of each PO.
- Students should think critically and reasonably.
- Students should be well aware of their competence at every level ofOBE.

Laboratory Code	BECL305	**CIE Marks**	50
Number of Lecture Hours/Week	02 Hours Laboratory	**SEE Marks**	50
Total Number of Lecture Hours	12	**Exam Hours**	03
CREDITS – 01			

CO-code	COURSE OUTCOMES	RBT
CO305.1	Design and analyze the BJT/FET amplifier and oscillator circuits.	L1, L2, L3
CO305.2	Design and test Opamp circuits to realize the mathematical computations, DAC and precision rectifiers.	
CO305.3	Design and test the combinational logic circuits for the given specifications.	
CO305.4	Test the sequential logic circuits for the given functionality.	
CO305.5	Demonstrate the basic electronic circuit experiments using SCR and 555 timer.	

	COURSE OBJECTIVE
CO1	Understand the electronic circuit schematic and its working.
CO2	Realize and test amplifier and oscillator circuits for the given specifications
CO3	Realize the opamp circuits for the applications such as DAC, implement mathematical functions and precision rectifiers.
CO4	Study the static characteristics of SCR and test the RC triggering circuit.
CO5	Design and test the combinational and sequential logic circuits for their functionalities.
CO6	Use the suitable ICs based on the specifications and functions

CO-PO-PSO Matrix

	PROGRAM OUTCOMES: PO												PSO	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	1	2
CO305.1	3	1						1	2	2	1	1	1	
CO305.2	3	2						1	2	1	1	1	1	
CO305.3	2	2						1	1	1	1	1	1	
CO305.4	2	2						1	1	1	1	1	1	
CO305.5	3	2						1	1	1	2	1	1	

GOVERNMENT ENGINEERING COLLEGE, RAICHUR-584135

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

CERTIFICATE

This is a controlled document of Department of Electronics and Communication Engineering, GEC Raichur-584135. The Document intended to be used as reference for students to carry out the laboratory experiments. This is prepared as per 2022 UG B.E Electronics and Communication scheme prescribed by VTU, Belagavi.

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Head of Dept.

Dept. of ECE

GEC, Raichur.

CONTENTS

PART-A: Hardware Experiments

1. Design and set up the BJT common emitter voltage amplifier with and without feedback and determine the gain- bandwidth product, input and output impedances.

2. Design and set-up BJT/FET i) Colpitts Oscillator, ii) Crystal Oscillator.

3. Design and set up the circuits using opamp: i) Adder, ii) Integrator, iii) Differentiator and iv) Comparator.

4. Design 4-bit R – 2R Op-Amp Digital to Analog Converter (i) for a 4-bit binary input using toggle switches (ii) by generating digital inputs using mod-16.

5. Design and implement (a) Half Adder & Full Adder using basic gates and NAND gates, (b) Half subtractor & Full subtractor using NAND gates, (c) 4-variable function using IC74151(8:1MUX).

6. Realize (i) Binary to Gray code conversion & vice-versa (IC74139), (ii) BCD to Excess-3 code conversion and vice versa.

7. a) Realize using NAND Gates: i) Master-Slave JK Flip-Flop, ii) D Flip-Flop and iii) T Flip-Flop

   b) Realize the shift registers using IC7474/7495: (i) SISO (ii) SIPO (iii) PISO (iv) PIPO (v) Ring counter and (vi) Johnson counter.

8. Realize a) Design Mod – N Synchronous Up Counter & Down Counter using 7476 JK Flip-flop b) Mod-N Counter using IC7490 / 7476 c) Synchronous counter using IC74192.

Demonstration Experiments (For CIE)

1. Design and Test the second order Active Filters and plot the frequency response,
   i) Low pass and Highpass Filter ii) Bandpass and Bandstop Filter.

2. Design and test the following using 555 timer i) Monostable ii) Astable multivibrator.

3. Design and Test regulated power supply.

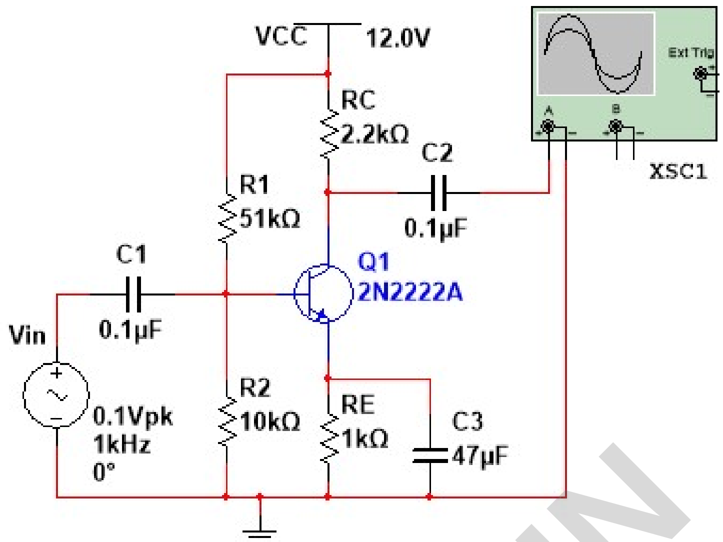
4. Design and Test an audio amplifier using by connecting a microphone input and observe the output using a loud speaker.

HARDWARE EXPERIMENTS

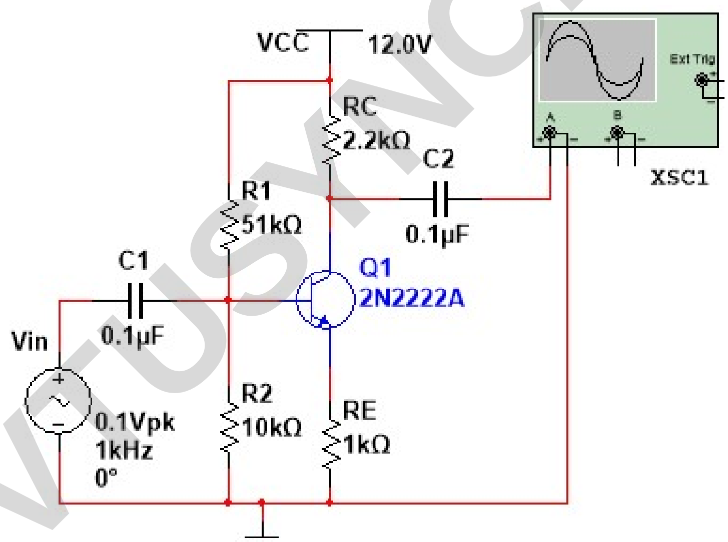
EXPERIMENT: 1

Common Emitter Amplifier

CIRCUIT DIAGRAM: Without Feedback



With Feedback



Design: Output requirement: Mid band v/g gain = 50 & required o/p v/g swing = 10V.

Transistor BC107 (h_{FE}> 100) $A_V = - h_{FE} (R_L / R_i)$, if $R_L = R_i$ then $A_V = - h_{FE}$

For BC107 I_C = 10mA [for CL100, I_C = 150mA] from datasheet

It is bias current at which h_{FE} measured.;

V_{CC}=12V [V_{CC} should be 20% more than o/p v/g swing] ,

$R_E = 0.1(V_{CC}/I_C) = 600\Omega$ [$V_{RE}/ I_E = V_{RE} / I_C$ since $I_E= I_C$ $R_E = V_{CC}$ x 10% / I_C]

R_C=0.4R_E = 2.4kΩ [since 40% V_{CC} is dropped across R_C]

$R_2 = V_{R2}/9I_B = 10k\Omega$; $R_1 = V_{R1}/ 10I_B = 51k\Omega$

[$I_B = I_C/ h_{FE}$ at least 10 times I_B should flow through R_1& R_2:

$V_{R1}= R_1 10I_B$, $V_{R2} = R_2 9I_B$ ::: $V_{R2} = V_{BE}+V_{RE}$ = (0.6+1.2) = 1.8V,

$V_{R1} = V_{CC} – V_{R2}$=(12-1.8)=10.2V]

C_E : = 0.26µF [choose f = 100Hz] X_{CE}<= R_E /10

$C_1 = 1/(2\pi f\ Ri)$ [$Ri = R1 \| R2 \| (1+ h_{FE})\ r_e$ :: where $r_e = 12.5\Omega$]

AIM: Design and set up the BJT common emitter amplifier with and without feedback and determine the gainbandwidthproduct from its frequency response.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1V to 50V and 1Hz to 30MHz	1
2.	Signal Generator	0.1V to 20V and 1Hz to 3MHz	1
2.	Transistor	BC107	1
3.	Power Supply	10V	1
4.	Resistors	680Ω, 2.4kΩ, 10kΩ, 51kΩ,	4
5.	Capacitors	0.22 µF, 0.01 µF	2,1

THEORY:

The single stage common emitter amplifier circuit commonly "Voltage Divider Biasing" The output voltage corresponding to all possible combinations of binary inputs is observed. This type of biasing arrangement uses two resistors as a potential divider network across the supply with their center point supplying the required Base bias voltage to the transistor. Voltage divider biasing is commonly used in the design of bipolar transistor amplifier circuits. This method of biasing the transistor greatly reduces the effects of varying Beta, (β) by holding the Base bias at a constant steady voltage level allowing for best stability. In **Common Emitter Amplifier** circuits, capacitors are used as **Coupling Capacitors** to separate the AC signals from the DC biasing voltage. This ensures that the bias condition set up for the circuit to operate correctly is not effected by any additional amplifier stages, as the capacitors will only pass AC signals and block any DC component. The **Voltage Gain** of the common emitter amplifier is equal to the ratio of the change in the input voltage to the change in the amplifiers output voltage. Then ΔV_L is Vout and ΔV_B is Vin.

PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Check the Biasing conditions of the Transistor Q1 by measuring $V_{CE} = V_C - V_E$.
3. Apply the V_{IN} signal at base junction of Q1 through Capacitor C1.
4. Note down the V_{IN} at set it to constant value, start vary the frequency and note down corresponding V_0 at collector terminal.
5. Calculate the Gain, Gain in dB and plot the frequency plot using semilog graph.
6. Compute the Bandwidth from the graph sheet, measure the gain bandwidth product.

Frequency Response without feedback (with C_E):

V_{IN} = ______V

Frequecny f in Hz	Output V_0 in Volts	Gain $A_V = (V_0/V_{IN})$	Gain in dB = $20\log(A_V)$

Frequency Response with feedback (without C_E):

V_{IN} = ______V

Frequecny f in Hz	Output V_0 in Volts	Gain $A_V = (V_0/V_{IN})$	Gain in dB = $20\log(A_V)$

7. Repeat the procedure for the common emitter with feedback circuit.

8. To Find Input Impedence:

   i. Connect potentiometer in between input and amplifier circuit.
   ii. Set the input frequency at mid gain frequency.
   iii. Vary the potentiometer till the mid gain frequency voltage reduced to half the signal strength.
   iv. Measure the potentiometer value to get input impedence.

9. Output Impedence:

   i. Connect potentiometer in parallel at output side of amplifier circuit,

   ii. Connect potentiometer in between input and amplifier circuit.

   iii. Set the input frequency at mid gain frequency.

   iv. Vary the potentiometer till the mid gain frequency voltage reduced to half the signal strength.

   v. Measure the potentiometer value to get input impedence.

RESULTS:

Common Emitter amplifier

	Without Feedback	**With Feedback**
GAIN		
BandWidth		

Pre Lab Questions:

1.What is an amplifier?

2. What are the classification of amplifiers?

3. What is input and output impedentce?

4. What is feedback?

Assignment Questions:

1. Define what is transistor biasing?

2. What are the efficiencies of power amplifier?

3. What are the Different feedback in amplifier?

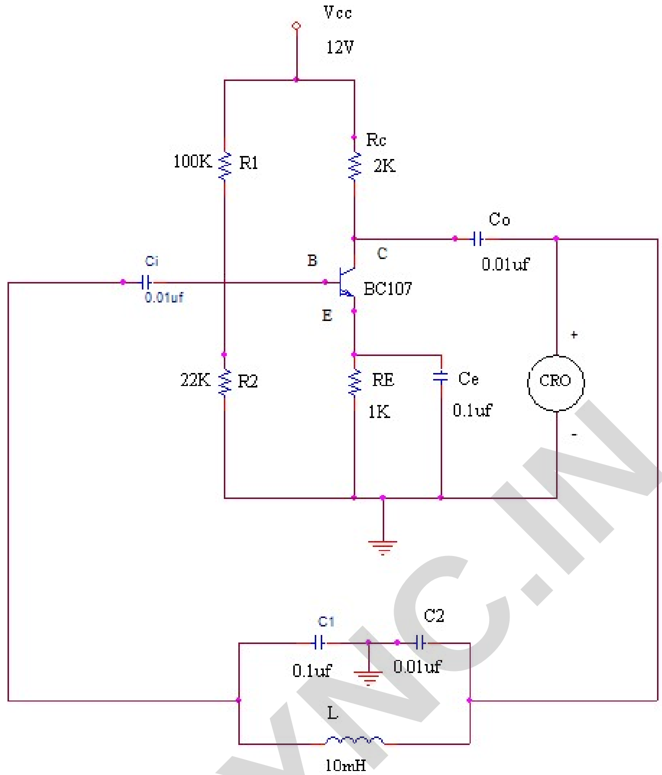
4. What is -3dB line in frequency response?

EXPERIMENT: 2

Oscillators

CIRCUIT DIAGRAM:

Colpitt Oscillator:



DESIGN:

Colpitts: [I] Assume f = 455kHz
$f = 1 / [2\pi(\sqrt{LC_{eq}})]$

$C_{eq} = C_1C_2/ (C_1+C_2)$ $A = C_2/C_1$ & $\beta = C1/C2$ the h_{FE} should be greater than (C_2/C_1) for sustained oscillations

$C_1=C_2=0.01\mu F$ then $L = 24.4\mu H$ [use $22\mu H$]

[II] **f=118kHz**

$C_1=0.1\ \mu F$, $C_2=0.01\ \mu F$, $L = 200\mu H$

R1= 100k, R2 = 10k, Rc = 2.2k, Re = 100, Cc =Cb= 0.01μF

AIM: Design and set-up BJT/FET, (a) Colpitts Oscillator (b) Crystal Oscillator.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1V to 50V and 1Hz to 30MHz	1
2.	Power Supply	0-30V	1
3.	Transistor	BC107	1
4.	Resistors	2.2k, 100, 10k, 100k	1 each
5.	Capacitors	0.01 µF, 0.1µF, 47µF	3,1, 1
6.	Inductors	200 µH	1

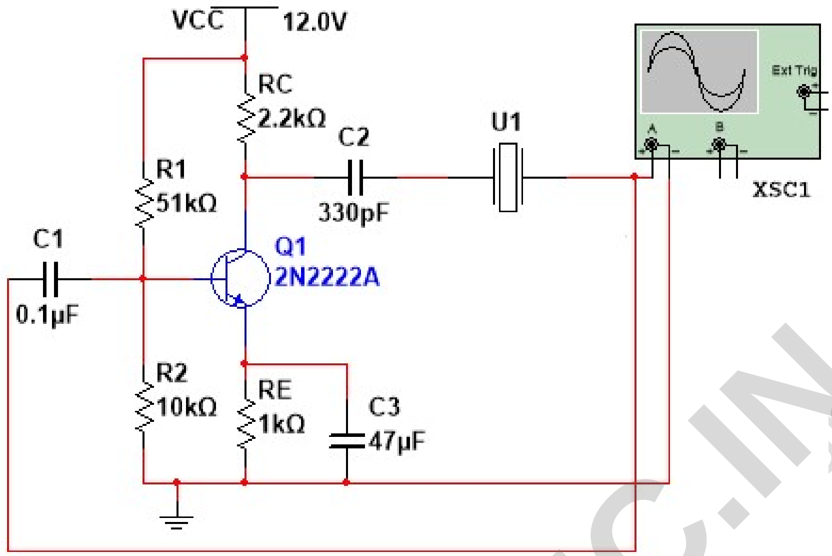
THEORY:

Colpitt's oscillator is very popular and is commonly used as local oscillator in radio receivers. The collector voltage is applied to the collector through inductor L whose reactance is high compared with X_2 and may therefore be omitted from equivalent circuit, at zero frequency; The circuit operates as Class C. the tuned circuit determines basically the frequency of oscillation.

PROCEDURE:

1. Setup the amplifier part of Colpitt's oscillator, check dc conditions. Ensure circuit works as an amplifier by giving an input.
2. Complete the circuit by connecting feedback circuit and observe output sinusoidal waveform, measure its amplitude and frequency.

CIRCUIT DIAGRAM:



DESIGN:

Requirement: Sine wave with amplitude 10V and frequency ____kHz.

DC bias conditions V_{CC} = 12V, I_C= 2mA from datasheet of transistor V_{RC} = 40%V_{CC}=4.8V, V_{RE} = 10%V_{CC}=1.2V, V_{CE} = 50%V_{CC}=6V

R_C = 2.2kΩ $R_C = V_{RC}/I_C$

R_E = 100Ω $R_E = V_{RE}/I_E$, $I_E=I_C$

R_2 = 10kΩ $V_{R2} = V_{BE}+V_{RE}$, $V_{R2} = 9I_BR_2$ $I_B = I_C/h_{FE}$,

R_1 = 100kΩ $V_{R1} = V_{CC} – V_{R1}$, $V_{R1} = 10I_BR_1$

Cc = 330 pF

Aim: Design the crystal oscillator.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1v to 80v and 1Hz to 260GHz	1
2.	Power Supply	0-30V	1
3.	Transistor	BC107	1
4.	Resisters	1k, 10k, 4.7k, 720	1 each
5.	Capacitors	330pF	1
6.	Crystal	5kHz	1

THEORY:

The conventional radio frequency oscillators using LC circuits, the frequency stability is usually poor because of the variations in temperature, humidity, transistor and circuit parameters, etc. for certain applications such as radio or television transmitters, it is essential that the frequency of oscillation of master oscillator must be extremely stable. The crystal oscillator is an excellent solution for this. It provides very high stability and quality factor. A piezoelectric crystal is used in crystal oscillator as a resonant tank circuit. A crystal acts like a large inductor it vibrates at the frequency of applied voltage. Conversely, if it is forced to vibrate, it will generate an alternating voltage. This property is called piezoelectric effect. The common piezoelectric crystals are Rochelle salt, Tourmaline and quartz. Since quartz is inexpensive and readily available compared to other crystals.

PROCEDURE:

1. Set up the amplifier part of the circuit and verify that it works as an amplifier
2. Complete the circuit connections and observe the output waveform.

RESULT:

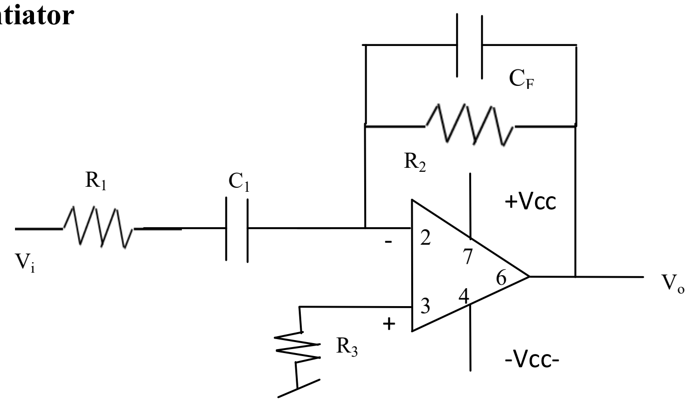
	Colpitts Oscillator	Crystal Oscillator
Theoretical :		
Practical :		

Oscillator is designed and verified for the given Oscillation frequency ________ Hz

EXPERIMENT: 3

Differentiator, Integrator, Adder and Comparator.

CIRCUIT DIAGRAM:
Differentiator



$$\boldsymbol{V_o = -R_F C_1 \frac{dV_i}{dt}}$$

Design Steps for Differentiator:

$$f_a = \frac{1}{2\pi R_2 C_1}$$

$$f_b = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi R_2 C_F}$$

f_c = unity gain bandwidth
T ≥ R_2 C_1

Select f_a equal to highest frequency of input signal to be differentiated.
Assume C_1 = 0.1µF , $f_a = f_{max} = 1.59kHz$

$$R_2 = \frac{1}{2\pi f_a C_1} = \frac{1}{2\pi\ 1.59k\ 0.1\mu} = 1k\Omega$$

Choose f_b = 20 f_aFind R_1 and C_F such that $R_1C_1 = R_2 C_F$
Since f_a = 1kHz ➔ f_b = 20 f_a= 20 (1k) = 20kHz.

$$R_1 = \frac{1}{2\pi f_b C_1} = \frac{1}{2\pi\ 20k\ 0.1\mu} = 79.57\ \Omega$$

$R_1C_1 = R_2 C_F$ ➔ $C_F = [R_1C_1] / R_2$ = (79.57) (0.01µ) / (1k) = 0.795nF

AIM: Design and setup the circuits using op-amp i) Adder ii) Integrator iii) Differentiator and iv) Comparator.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1V to 80V and 1Hz to 260GHz	1
2.	Voltmeter	0 to 20 V	1
3.	Power Supply	+/- 12v , 5V	1, 1
4.	Op-Amp	741 +/- Vsat = 15v	1
5.	Resistor	79Ω, 1.59k, 10kΩ	1,1,1
6.	Capacitors	0.01µF	2

THEORY:

An op-amp differentiator or a differentiating amplifier is a circuit configuration which produces output voltage amplitude that is proportional to the rate of change of the applied input voltage. An op-amp differentiator is an inverting amplifier, which uses a capacitor in series with the input voltage. Differentiating circuits are usually designed to respond for triangular and rectangular input waveforms. For a sine wave input, the output of a differentiator is also a sine wave, which is out of phase by Φ with respect to the input (cosine wave).

An op-amp integrating circuit produces an output voltage which is proportional to the area (amplitude multiplied by time) contained under the waveform. An ideal op-amp integrator uses a capacitor C_1, connected between the output and the op-amp inverting input terminal, square wave, the change in the input signal amplitude charges and discharges the feedback capacitor. This results in a triangular wave output with a frequency that is dependent on the value of (R_1C_f), which is referred to as the time constant of the circuit. It provided with a sine wave input whose frequency is varying, the integrator behaves like a "Low-Pass Filter", which produces only low frequency signal at the output. All the high frequency signal components are blocked or attenuated.

Summing amplifier is also called as Voltage adder since its output is the addition of voltages present at its input terminal. The summing amplifier uses an inverting amplifier configuration. The inverting amplifier circuit has only one voltage at the inverting input terminal. If more input voltages are connected to the inverting input terminal, the resulting output will be the sum of all the input voltages applied, but inverted. The gain for each input is given by the ratio of the feedback resistor R_f to the input resistance in the respective branch.

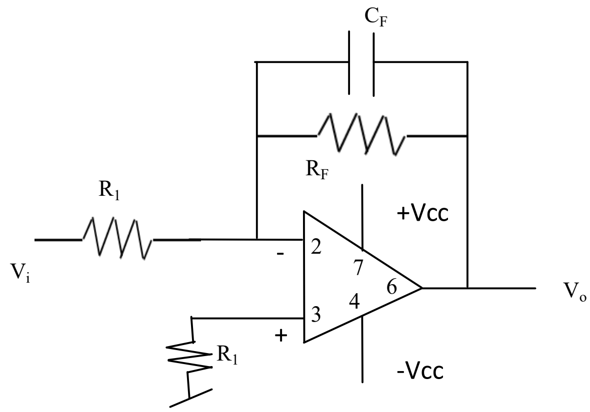
PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Adjust the Resistor R and C and observe the output waveforms of the timer.

RESULT:

Working of Integrator, Differentiator, Adder Verified

Integrator:



$$V_o = -\frac{1}{R_F C_1}\int V_i\,dt$$

Design Steps for Integrator:

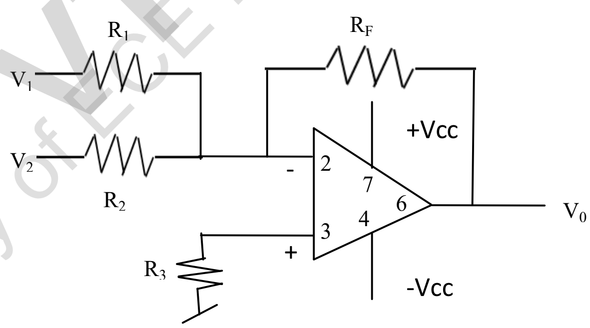
$f_a = 1 / [2\pi R_F C_F]$ After f_a gain decreases at rate of 20dB/decade.
$f_b = 1 / [2\pi R_1 C_F]$ Frequency at which gain is 0dB
Generally: $f_a < f_b$ ➔ If $f_a = f_b/10$ *then* $R_F = 10 R_1$
$T \geq R_F C_F$ ➔ $R_F C_F = 1/2\pi f_a$

Given $f_a = 1\ kHz$, therefore $T = 1msec$

Assume $C_F = 0.01\mu F$, then $R_F = 1msec / 0.01\mu F = 100K\Omega$

Then, $R_1 = R_F / 10$ ➔ $R_1 = 10K\Omega$

Adder:



$$V_o = -\left\{\frac{R_F}{R_1}V_1 + \frac{R_F}{R_2}V_2\right\}$$

Adder:
To work Inverting Adder to produce $V_0 = -(V_1 + V_2)$
$(R_f/R_1) = 1$ and $(R_f/R_2) = 1$ therefore $R_1 = R_2 = R_f = R = 1k\Omega$

OBSERVATION:

Integration	Vin= ____V	Freq = ____Hz	Vo= ____V	Freq = ____Hz
Differentiation	Vin= ____V	Freq = ____Hz	Vo= ____V	Freq = ____Hz
Adder	V_1= 1V V_2= __V (vary)	Vo=____V	V_2= 1V V_1= __V (vary)	Vo=____V

Pre Lab Questions:

1. What is Integration circuit? What is Differentiation circuit?
2. What is summing amplifier?
3. What is comparator?
4. Define the gain of inverting & non-inverting Amplifiers?

Lab Assignment Questions:

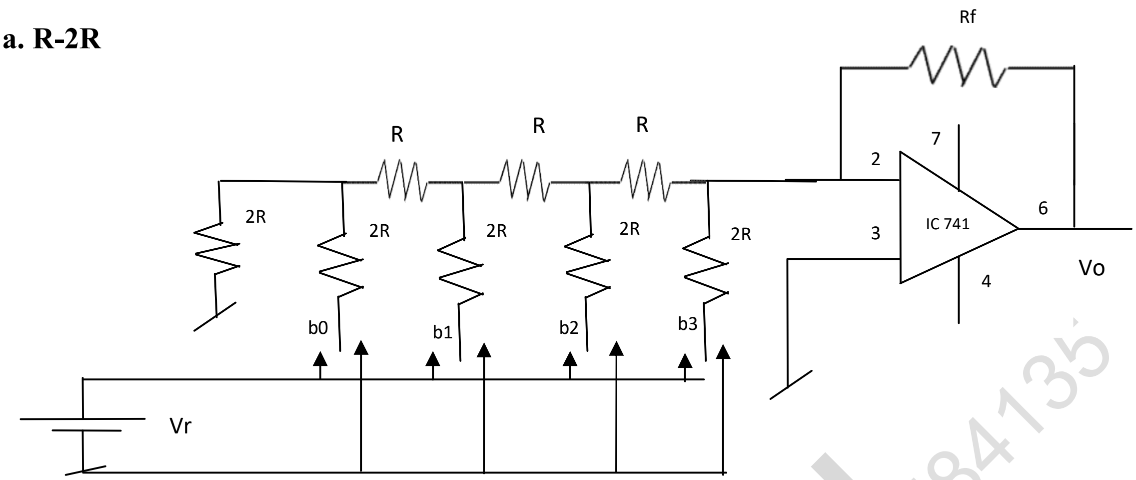
1. How to calculate Phase shift in Integrator & Differentiator?
2. Is it feasible to design summing amplifier with unequal gain?
3. Type of feedback used in comparator circuits?
4. Nature of output that can be generated from comparator circuits?
5. Differentiate Comparator and Oscillator circuits.

EXPERIMENT No.: 4

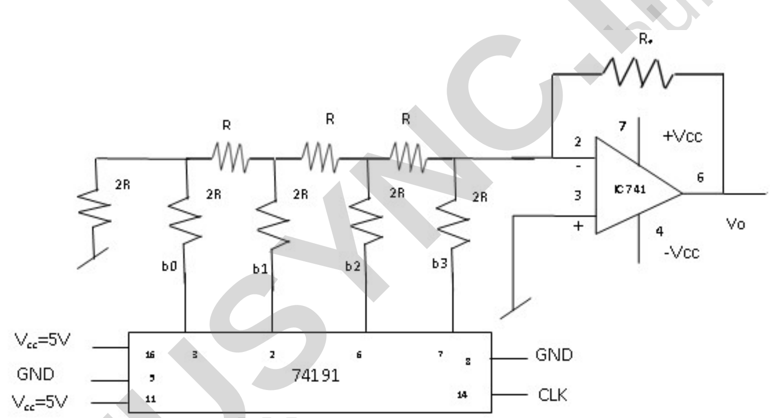
DAC using R-2R Resistor Network

CIRCUIT DIAGRAM:

a. R-2R



b. R-2R inputs mod16 counter



READINGS IN TABULAR COLUMN:

Decimal equivalent binary inputs	Input (V)				Theoratical voltage (Vo) When R_F=2R,, V_r = 5V	Observed Output Voltage (Vo)
	b_3	b_2	b_1	b_0		
0	0	0	0	0	0	
1	0	0	0	1	-0.625	
2	0	0	1	0	-1. 25	
.	.	.	.	.	.	
.	.	.	.	.	.	
15	1	1	1	1	-9.735	

AIM: Design DAC using R-2R resistor network.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1v to 80v and 1Hz to 260GHz	1
2.	Voltmeter	0 to 20 V	1
3.	Power Supply	+/- 12v , 5V	1, 1
4.	Op-Amp, Binary counter	741 +/- Vsat = 15v, IC74191	1
5.	Resistor	5k, 10k	3,6

THEORY:

D/A converter with R and 2R resistors, the binary inputs are simulated by switches b_0 through b_3, and the output is proportional to the binary inputs. Binary inputs can be in the high (+5V) or low(0V) state.

The output voltage corresponding to all possible combinations of binary inputs is observed.

PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Apply the V_r as $5V$ and $0V$ to various combinations of inputs b_0, b_1, b_2, b_3.
3. Note down the corresponding output voltage.

DESIGN:

Assume the R = 10 kΩ , corresponding 2R = 20 kΩ

1. Let $V_r = 5V$, $V_o = -V_r\left(\frac{R_F}{R}\right)\left(\frac{b_3}{2^1}+\frac{b_2}{2^2}+\frac{b_1}{2^3}+\frac{b_0}{2^4}\right)$
2. $R_{TH} = 2R$, then current through 2R connected to +5V is ($5v/R_{TH}\Omega$) = 1 A
   Same current flows through R_f and output will be
   $V_o = -(R_f\Omega) \times I$ A = - V_r V Therefore R_f = 2R Ω

RESULTS:

The R-2R circuit is designed for 4-bits and verified for DAC operation.

Pre Lab Questions:

1. What is DAC?
2. R-2R ladder necessity?

Lab Assignment Questions:

1. What are applications of DAC?
2. Different types of DAC?
3. How to Design n-bit DAC circuit?
4. What is the step size of DAC?

EXPERIMENT: 5

Half and Full Adder

Design: (a) Half Adder & Full Adder

Half Adder				Full Adder				
A	**B**	**Carry**	**Sum**	**A**	**B**	**C**	**Carry**	**Sum**
0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	1	0	1
1	0	0	1	0	1	0	0	1
1	1	1	0	0	1	1	1	0
				1	0	0	0	1
$Sum = A \oplus B$ $Carry = AB$				1	0	1	1	0
				1	1	0	1	0
				1	1	1	1	1

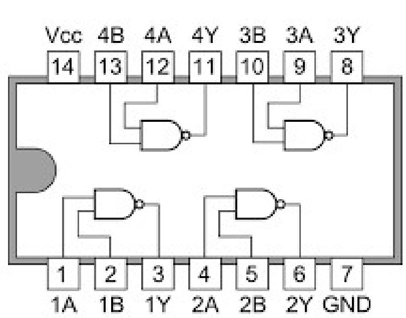
$$Carry = AC + AB + BC = A(B + C) + BC$$

A \ BC	**00**	**01**	**11**	**10**
0	0	0	1	0
1	0	1	1	1

$$Sum = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC = A(\overline{B}\overline{C} + BC) + \overline{A}(\overline{B}C + B\overline{C})$$
$$= A(\overline{B \oplus C}) + \overline{A}(B \oplus C) = A \oplus B \oplus C$$

A \ BC	**00**	**01**	**11**	**10**
0	0	1	0	1
1	1	0	1	0

7400 Quad 2-input NAND Gates



Aim: Design and implement (a) Half Adder & Full Adder using basic gates and NAND gates, (b) Half subtractor & Full subtractor using NAND gates, (c) 4-variable function using IC74151(8:1MUX).

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	Logic Gates	7404, 7432, 7408	2 each
2.	Power Supply	+ 5V	1
3.	LED with resistor	1kΩ	2 each

THEORY:

A combinational logic circuit which is designed to add two binary digits is known as half adder. The half adder provides the output along with a carry value. A combinational circuit which is designed to add three binary digits and produce two outputs is known as full adder. The full adder circuit adds three binary digits, where two are the inputs and one is the carry forwarded from the previous addition.

Half Subtractors are a type of digital circuit that calculates the arithmetic binary subtraction between two single-bit numbers. It is a circuit with two inputs and two outputs.A full Subtractor is a digital circuit that performs the subtraction of three single-digit binary numbers. This is a three-input and two-output digital circuit.

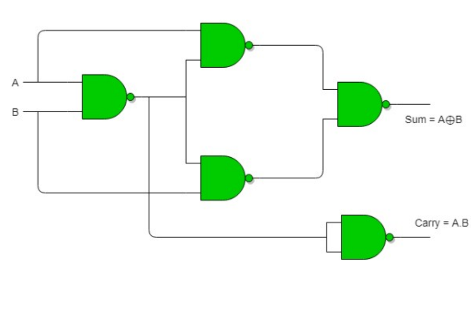
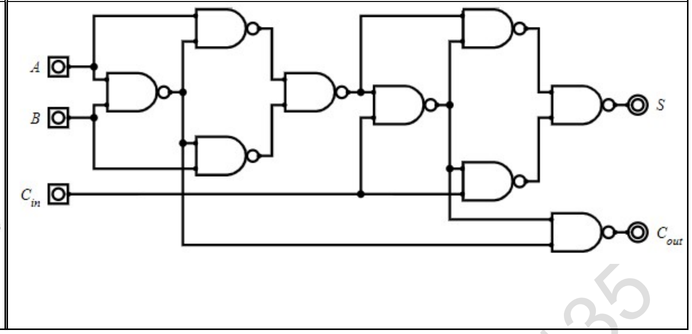
A multiplexer (MUX) is a digital combinational circuit (logic circuit). A multiplexer has multiple inputs and a single output. The output signal depends upon the input signal given to the select lines.A Multiplexer (MUX) that has 8 inputs and a single output is known as 8:1 Multiplexer.

Procedure:

1. Connections are made as shown in circuit diagram
2. Verify the output for Half Adder and Full Adder according to truth table.
3. Verify the output for Half Subtractor and Full Subtractor according to truth table.

Results:

The working of Half and Full Adder verified using basic gates and NAND gates.

Half Adder using NAND	Full Adder using NAND
	

Design: (b) Half subtractor & Full Subtractor

Half Subtractor				Full Subtractor				
A	**B**	**Diff**	**Barrow**	**A**	**B**	**C**	**Diff**	**Barrow**
0	0	0	0	0	0	0	0	0
0	1	1	1	0	0	1	1	1
1	0	1	0	0	1	0	1	1
1	1	0	0	0	1	1	0	1
				1	0	0	1	0
				1	0	1	0	0
				1	1	0	0	0
				1	1	1	1	1

$$Diff = A \oplus B$$
$$Barrow = \overline{A}B$$

Difference:

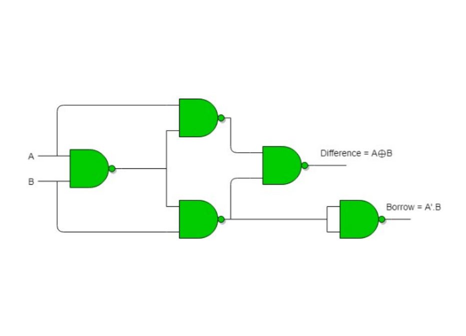
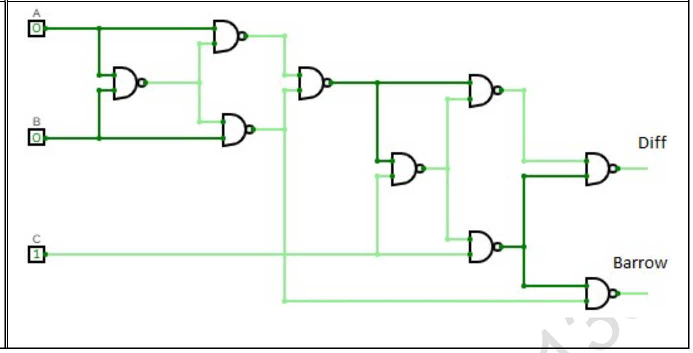
A \ BC	**00**	**01**	**11**	**10**
0	0	1	0	1
1	1	0	1	0

$$Diff = \overline{A}\,\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\,\overline{C} + ABC = A(\overline{B}\,\overline{C} + BC) + \overline{A}(\overline{B}C + B\overline{C})$$
$$= A(B \oplus C) + \overline{A}\left(\overline{B \oplus C}\right) = A \oplus B \oplus C$$

Barrow:

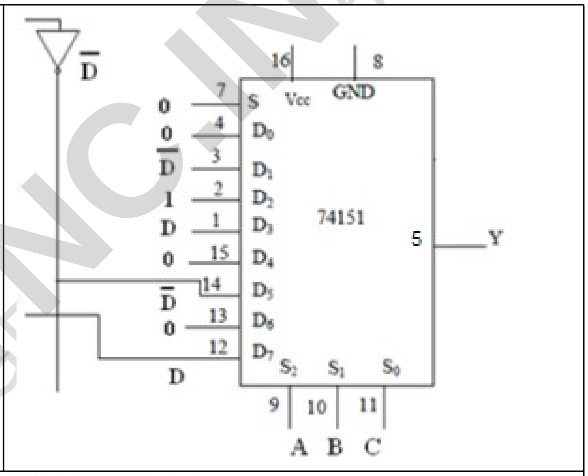
A \ BC	**00**	**01**	**11**	**10**
0	0	1	1	1
1	0	0	1	0

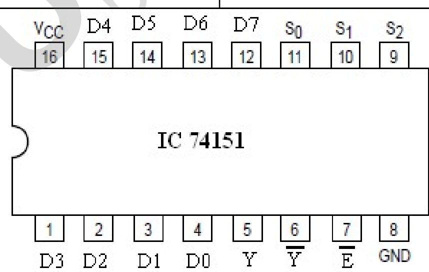
$$Barrow = \overline{A}C + \overline{A}B + BC = \overline{A}(B + C) + BC$$

Half Subtractor using NAND	Full Subtractor using NAND
	

(c) 4-variable function using IC74151(8:1MUX): $f(A,B,C,D) = \sum m(2,4,5,7,10,14)$

DC \ BA	00	01	11	10
00				1
01	1	1	1	
11				1
10				1





EXPERIMENT: 6

Code Conversion

Truth Table:

Binary				**Grey**				**Grey**				**Binary**			
B3	**B2**	**B1`**	**B0**	**G3**	**G2**	**G1`**	**G0**	**G3**	**G2**	**G1`**	**G0**	**B3**	**B2**	**B1`**	**B0**
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1	0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0	0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0	0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	1	0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	1	0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	0	0	1	1	1	0	1	0	1
1	0	0	0	1	1	0	0	1	0	0	0	1	1	1	1
1	0	0	1	1	1	0	1	1	0	0	1	1	1	1	0
1	0	1	0	1	1	1	1	1	0	1	0	1	1	0	0
1	0	1	1	1	1	1	0	1	0	1	1	1	1	0	1
1	1	0	0	1	0	1	0	1	1	0	0	1	0	0	0
1	1	0	1	1	0	1	1	1	1	0	1	1	0	0	1
1	1	1	0	1	0	0	1	1	1	1	0	1	0	1	1
1	1	1	1	1	0	0	0	1	1	1	1	1	0	1	0

Binary to Grey conversion

$g_0=b_0\oplus b_1$, $g_1=b_1\oplus b_2$, $g_2=b_1\oplus b_2$, …. $g_{(n-1)}=b_{(n-1)}\oplus b_n$

Aim: Realize (i) Binary to Gray code conversion & vice-versa (IC74139), (ii) BCD to Excess-3 code conversion and vice versa.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	Logic Gates	7486, 7483, 7404, 7408, 7432	2 each
2.	Power Supply	+ 5V	1
3.	LED with resistor	1kΩ	4 each

THEORY:

Gray code – also known as Cyclic Code, Reflected Binary Code (RBC), Reflected Binary (RB) or Grey code – is defined as an ordering of the binary number system such that each incremental value can only differ by one bit. In gray code, while traversing from one step to another step only one bit in the code group changes. That is to say that two adjacent code numbers differ from each other by only one bit.

Gray code is the most popular of the unit distance codes, but it is not suitable for arithmetic operations. Gray code has some applications in analog to digital converters, as well as being used for error correction in digital communication.

Binary to Gray conversion:

a. The Most Significant Bit (MSB) of the gray code is always equal to the MSB of the given binary code.

b. Other bits of the output gray code can be obtained by XORing binary code bit at that index and previous index.

Gray to binary conversion:

a. The Most Significant Bit (MSB) of the binary code is always equal to the MSB of the given gray code.

b. Other bits of the output binary code can be obtained by checking the gray code bit at that index. If the current gray code bit is 0, then copy the previous binary code bit, else copy the invert of the previous binary code bit.

Procedure:

1. Connections are made as shown in circuit diagram
2. Verifty the output for Half Adder and Full Adder according to truth table.

Results:

The working of Code conversion form Binary to Grey and vice versa observed as per the truth table.

K-MAP: Binary to Grey

G0

B3B2 \ B1B0	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

G1

B3B2 \ B1B0	00	01	11	10
00			1	1
01	1	1		
11	1	1		
10			1	1

G2

B3B2 \ B1B0	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

G3

B3B2 \ B1B0	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10				

Grey to Binary

B0

G3G2 \ G1G0	00	01	11	10
00	0	1		1
01	1		1	
11		1		1
10	1		1	

B1

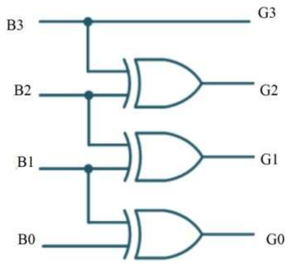
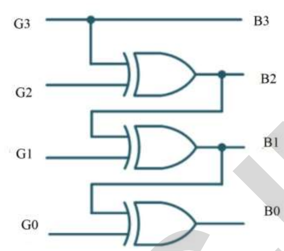
G3G2 \ G1G0	00	01	11	10
00			1	1
01	1	1		
11			1	1
10	1	1		

B2

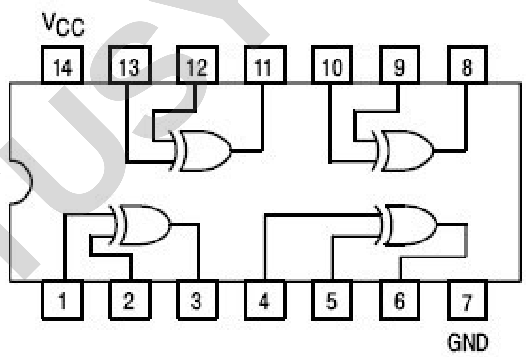
G3G2 \ G1G0	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

B3

G3G2 \ G1G0	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

Binary to Gray	Gray to Binary
$G_0 = B_0 \oplus B_1$ $G_1 = B_1 \oplus B_2$ $G_2 = B_2 \oplus B_3$ $G_3 = B_3$	$B_0 = G_0 \oplus G_1 \oplus G_2 \oplus G_3$ $B_1 = G_1 \oplus G_2 \oplus G_3$ $B_2 = G_3 \oplus G_2$ $B_3 = G_3$
	

7486 Pin Diagram



Truth Table: BCD to Excess-3

BCD				**Excess-3**				**Excess-3**				**BCD**			
B_3	**B_2**	**B_1`**	**B_0**	**E_3**	**E_2**	**E_1`**	**E_0**	**E_3**	**E_2**	**E_1`**	**E_0**	**B_3**	**B_2**	**B_1`**	**B_0**
0	0	0	0	0	0	1	1	0	0	0	0	X	X	X	X
0	0	0	1	0	1	0	0	0	0	0	1	X	X	X	X
0	0	1	0	0	1	0	1	0	0	1	0	X	X	X	X
0	0	1	1	0	1	1	0	0	0	1	1	0	0	0	0
0	1	0	0	0	1	1	1	0	1	0	0	0	0	0	1
0	1	0	1	1	0	0	0	0	1	0	1	0	0	1	0
0	1	1	0	1	0	0	1	0	1	1	0	0	0	1	1
0	1	1	1	1	0	1	0	0	1	1	1	0	1	0	0
1	0	0	0	1	0	1	1	1	0	0	0	0	1	0	1
1	0	0	1	1	1	0	0	1	0	0	1	0	1	1	0
1	0	1	0	X	X	X	X	1	0	1	0	0	1	1	1
1	0	1	1	X	X	X	X	1	0	1	1	1	0	0	0
1	1	0	0	X	X	X	X	1	1	0	0	1	0	0	1
1	1	0	1	X	X	X	X	1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X	1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X	1	1	1	1	X	X	X	X

K-MAP: BCD to Excess-3

E_0

B_3B_2 \ B_1B_0	00	01	11	10
00	1			1
01	1			1
11	X	X	X	X
10	1		X	X

E_1

B_3B_2 \ B_1B_0	00	01	11	10
00	1		1	
01	1		1	
11	X	X	X	X
10	1		X	X

E_2

B_3B_2 \ B_1B_0	00	01	11	10
00		1	1	1
01	1			
11	X	X	X	X
10		1	X	X

E_3

B_3B_2 \ B_1B_0	00	01	11	10
00				
01		1	1	1
11	X	X	X	X
10	1	1	X	X

Excess-3 to BCD

B_0

E_3E_2 \ E_1E_0	00	01	11	10
00	X	X		X
01	1			1
11	1	X	X	X
10	1			1

B_1

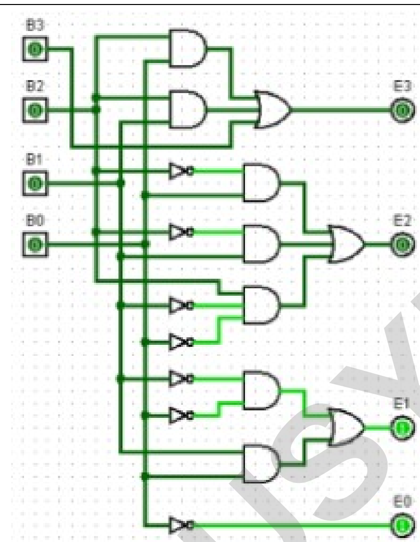
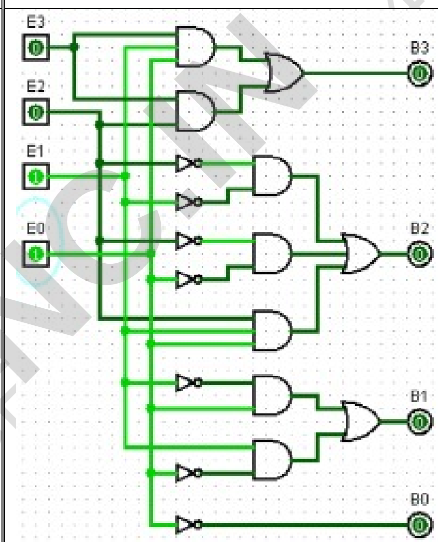
E_3E_2 \ E_1E_0	00	01	11	10
00	X	X		X
01		1		1
11		X	X	X
10		1		1

B_2

E_3E_2 \ E_1E_0	00	01	11	10
00	X	X		X
01			1	
11		X	X	X
10	1	1		1

B_3

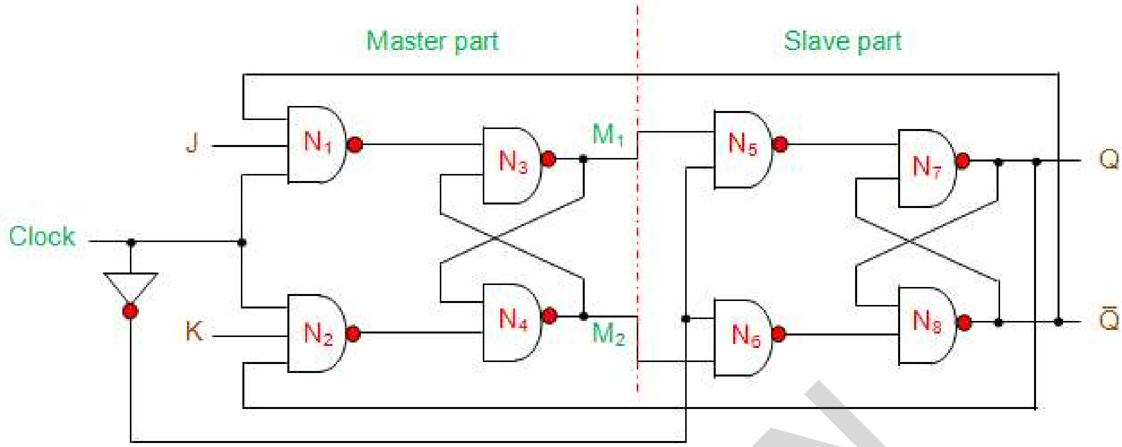
E_3E_2 \ E_1E_0	00	01	11	10
00	X	X		X
01				
11	1	X	X	X
10			1	

BCD to E-3	E-3 to BCD
$E_0 = \overline{B_0}$ $E_1 = \overline{B_0}\,\overline{B_1} + B_0B_1$ $= B_0 \oplus B_1$ $E_2 = \overline{B_2}B_0 + \overline{B_2}B_1 + \overline{B_0B_1}B_2$ $E_3 = B_0B_2 + B_1B_2 + \overline{B_1}B_3$	$B_0 = \overline{E_0}$ $B_1 = E_1\overline{E_0} + E_0\overline{E_1} = E_0 \oplus E_1$ $B_2 = \overline{E_1E_2} + E_0E_1E_2 + \overline{E_0E_2}$ $B_3 = E_0E_1E_3 + E_2E_3$
	

EXPERIMENT: 7

Flip Flop and Shift Registers

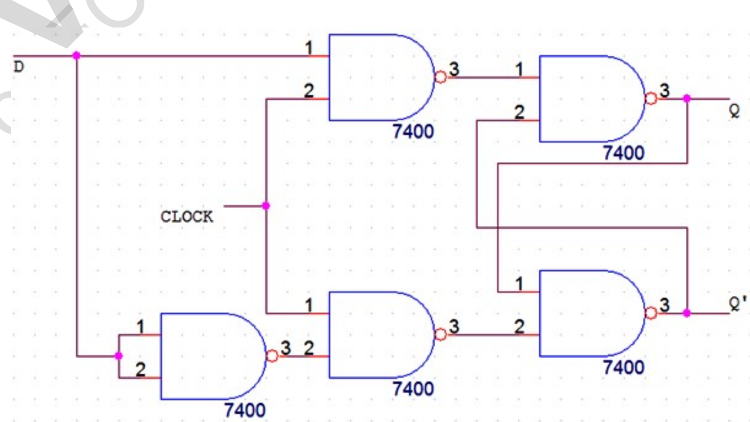
Circuit Diagram: a) Realize using NAND gate - MSJK FF, D-FF, T-FF



Truth Table:

Input		Present State		Next State		
J	K	Q	$\bar{Q}$	Q	$\bar{Q}$	
0	0	0	1	0	1	No Change
		1	0	1	0	
0	1	0	1	0	1	Reset
		1	0	0	1	
1	0	0	1	1	0	Set
		1	0	1	0	
1	1	0	1	1	0	Toggle
		1	0	0	1	

D-FlipFlop:



Aim: a) Realize using NAND Gates: i) Master-Slave JK Flip-Flop, ii) D Flip-Flop and iii) T Flip-Flop.

b) Realize the shift registers using IC7474/7495: (i) SISO (ii) SIPO (iii) PISO (iv) PIPO (v) Ring counter and (vi) Johnson counter.

Components & Apparatus:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	Logic Gates	7410, 7400, 7404, 7474, 7495,	2 each
2.	Power Supply	+ 5V	1
3.	LED with resistor	1kΩ	2 each

THEORY:

MS JK Flip Flop:

The master-slave flip flop is constructed by combining two JK flip flops. These flip flops are connected in a series configuration. In these two flip flops, the 1st flip flop work as "master", called the master flip flop, and the 2nd work as a "slave", called slave flip flop. The master-slave flip flop is designed in such a way that the output of the "master" flip flop is passed to both the inputs of the "slave" flip flop. The output of the "slave" flip flop is passed to inputs of the master flip flop.

Operation of MS JK Flip Flop:

- When the clock pulse is true, the slave flip flop will be in the isolated state, and the system's state may be affected by the J and K inputs. The "slave" remains isolated until the CP is 1. When the CP set to 0, the master flip-flop passes the information to the slave flip flop to obtain the output.
- The master flip flop responds first from the slave because the master flip flop is the positive level trigger, and the slave flip flop is the negative level trigger.
- The output Q'=1 of the master flip flop is passed to the slave flip flop as an input K when the input J set to 0 and K set to 1. The clock forces the slave flip flop to work as reset, and then the slave copies the master flip flop.
- When J=1, and K=0, the output Q=1 is passed to the J input of the slave. The clock's negative transition sets the slave and copies the master.
- The master flip flop toggles on the clock's positive transition when the inputs J and K set to 1. At that time, the slave flip flop toggles on the clock's negative transition.
- The flip flop will be disabled, and Q remains unchanged when both the inputs of the JK flip flop set to 0.

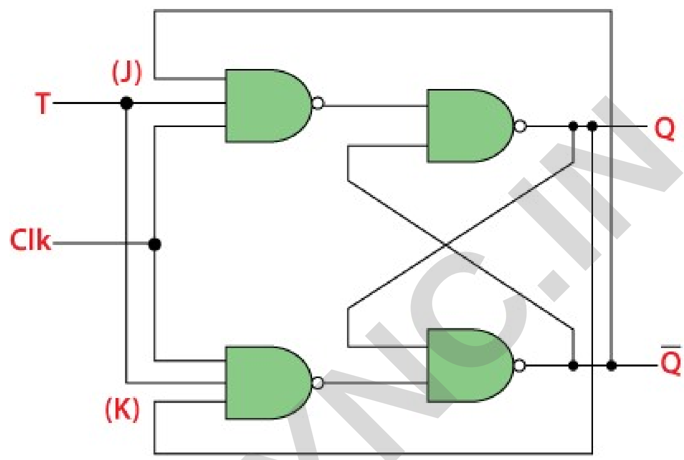
D-Flip Flop:

Whenever the clock signal is LOW, the input is never going to affect the output state**.** The clock has to be high for the inputs to get active. Thus, D flip-flop is a controlled Bi-stable latch where the clock signal is the control signal. Again, this gets divided into positive edge triggered D flip flop and negative edge triggered D flip-flop**.**

Truth Table:

D	Output	
	Q	$\bar{Q}$
0	0	1
1	1	0

T-Flip Flop:



Truth Table:

	Previous		Next	
T	Q	$\bar{Q}$	Q	$\bar{Q}$
0	0	1	0	1
0	1	0	1	0
1	0	1	1	0
1	1	0	0	1

Operations of T-Flip Flop

The next sate of the T flip flop is similar to the current state when the T input is set to false or 0.

- If toggle input is set to 0 and the present state is also 0, the next state will be 0.
- If toggle input is set to 0 and the present state is 1, the next state will be 1.

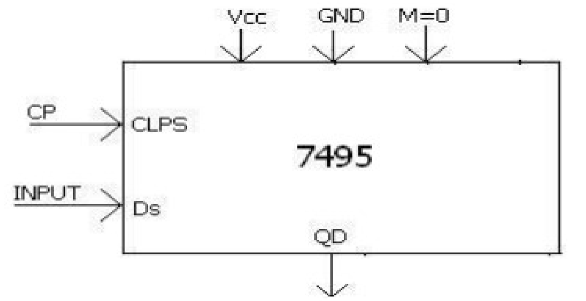
The next state of the flip flop is opposite to the current state when the toggle input is set to 1.

- If toggle input is set to 1 and the present state is 0, the next state will be 1.
- If toggle input is set to 1 and the present state is 1, the next state will be 0.

b) Realize the shift registers using IC7474/7495: (i) SISO (ii) SIPO (iii) PISO (iv) PIPO (v) Ring counter and (vi) Johnson counter.

(i) SERIAL IN SERIAL OUT (SISO)-shift right

CLK		Q_A	Q_B	Q_C	Q_D
0	1	1			
1	0	0	1		
2	1	1	0	1	
3	1	1	1	0	1
4		X	1	1	1
5		X	X	0	0
6		X	X	X	1

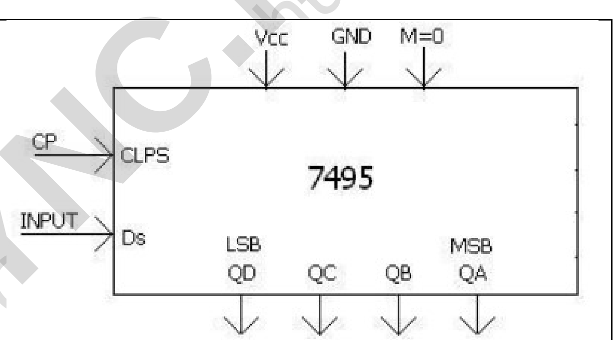


SISO:- 1. M=0, clks-> CP,

2. Input is given at DS[give 1 or 0 at DS press the mono pulsar]
3. The o/p shifts right Q_A, to Q_D.
4. After the 4th clock pulse o/p is seen at Q_A, Q_B, Q_C,Q_D.
5. Continue pressing the mono pulse , o/p is seen at Q_D.

(ii) SERIAL IN PARALLEL OUT (SIPO)

CLK		Q_A	Q_B	Q_C	Q_D
0	1	1			
1	0	0	1		
2	1	1	0	1	
3	1	1	1	0	1



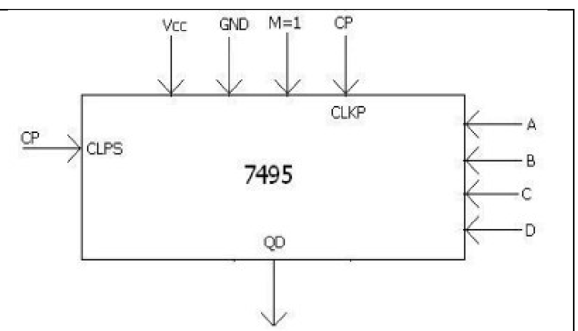
SIPO:-

After the 4th clock pulse o/p is seen at. Q_A, Q_B, Q_C,Q_D.

Example if i/p is 1011 o/p is 1101

(iii) PARALLEL IN SERIAL OUT (PISO)

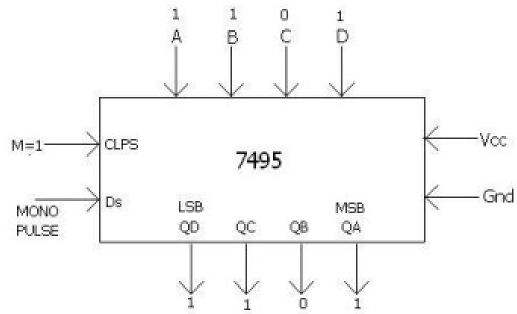
CLK		Q_A	Q_B	Q_C	Q_D
0	1	1	1	0	1
1	0	X	1	1	0
2	1	X	0	1	1
3	1	X	X	X	1



PISO:-

1. M=1, clk-> CP, clks- >1.
2. Load the parallel data ABCD , which gets stored in Q_A, Q_B, Q_C,Q_D.
3. Clks- >CP, M=0.
4. Output is observed at Q_D.

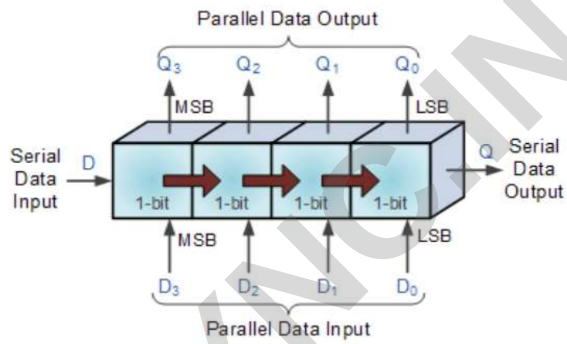
(iv) PARALLEL IN PARALLEL OUT (PIPO)



PIPO:-

1. M=1, clk-> CP.
2. Give the data through ABCD , give CP.
3. The o/p is stored inQ_A, Q_B, Q_C,Q_D.

Illustration of SISO, SIPO, PISO, and PIPO



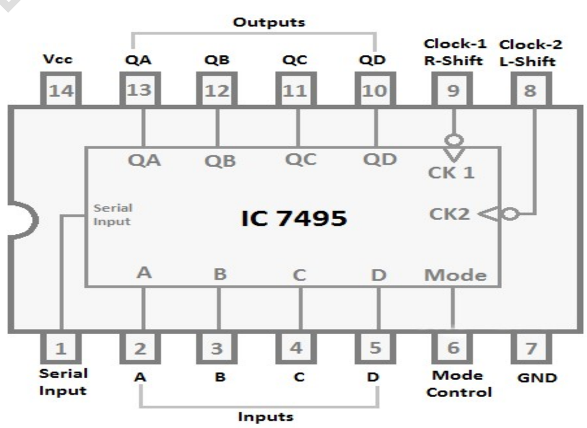
Procedure:

1. Connections are made as shown in circuit diagram
2. Verifty the output for MS JK, D, and T flip flop according to the truth table.

Results:

The working of MS JK, D, T Flip Flop working is verified according to truth table.
The working of SISO, SIPO, PISO, PIPO verified.

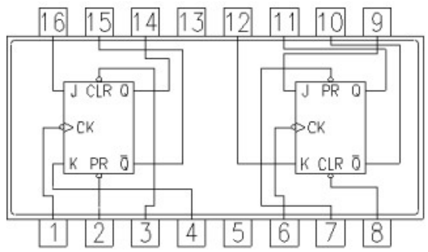
7495 Pin Diagram



EXPERIMENT: 8

MOD N Counters

Circuit Diagram:i) up Counter



Present State	Next State	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Present State			Next State			Input for Flip-flop					
Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Q_C \ Q_BQ_A	00	01	11	10
0	1	X	X	1
1	1	X	X	1

$J_A = 1$

Q_C \ Q_BQ_A	00	01	11	10
0	X	1	1	X
1	X	1	1	X

$K_A = 1$

Q_C \ Q_BQ_A	00	01	11	10
0		1	X	X
1		1	X	X

$J_B = Q_A$

Q_C \ Q_BQ_A	00	01	11	10
0	X	X	1	
1	X	X	1	

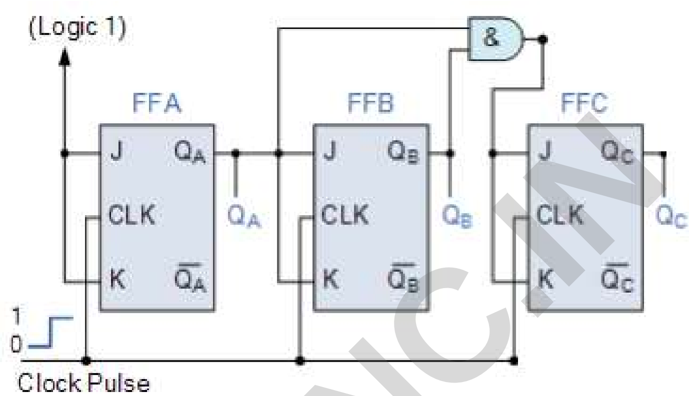
$K_B = Q_A$

Q_C \ Q_BQ_A	00	01	11	10
0			1	
1	X	X	X	X

$J_B = Q_AQ_B$

Q_C \ Q_BQ_A	00	01	11	10
0	X	X	1	
1	X	X	1	

$K_B = Q_AQ_B$



ii) Down Counter

Present State			Next State			Input for Flip-flop					
Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_C	K_C	J_B	K_B	J_A	K_A
0	**0**	**0**	1	1	1	**1**	**X**	1	X	**X**	**1**
0	**0**	**1**	0	0	0	**0**	**X**	0	X	**1**	**X**
0	**1**	**0**	0	0	1	**0**	**X**	X	1	**X**	**1**
0	**1**	**1**	0	1	0	**0**	**X**	X	0	**1**	**X**
1	**0**	**0**	0	1	1	**X**	**1**	1	X	**X**	**1**
1	**0**	**1**	1	0	0	**X**	**0**	0	X	**1**	**X**
1	**1**	**0**	1	0	1	**X**	**0**	X	1	**X**	**1**
1	**1**	**1**	1	1	0	**X**	**0**	X	0	**1**	**X**

Q_C \ Q_BQ_A	00	01	11	10
0	X	1	1	X
1	X	1	1	X

$J_A = 1$

Q_C \ Q_BQ_A	00	01	11	10
0	1	X	X	1
1	1	X	X	1

$K_A = 1$

Q_C \ Q_BQ_A	00	01	11	10
0	1		X	X
1	1		X	X

$J_B = \overline{Q_B}$

Q_C \ Q_BQ_A	00	01	11	10
0	X	X		1
1	X	X		1

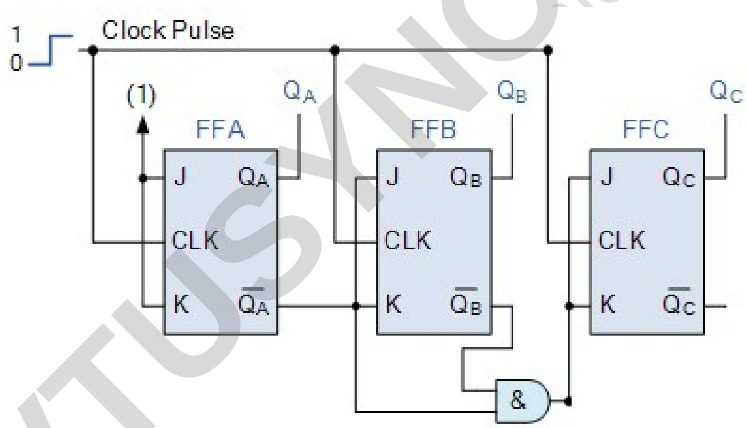
$K_B = \overline{Q_B}$

Q_C \ Q_BQ_A	00	01	11	10
0	1			
1	X	X	X	X

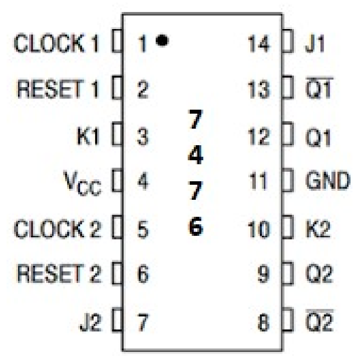
$J_C = \overline{Q_A Q_B}$

Q_C \ Q_BQ_A	00	01	11	10
0	X	X	X	X
1	1			

$K_B = \overline{Q_A Q_B}$



7476 Pin Diagram



Aim: Realize a) Design Mod N Synchronous Up Counter & Down Counter using 7476 JK Flip-flop b) Mod-N Counter using IC7490 / 7476 c) Synchronous counter using IC74192.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	Logic Gates	7476, 7490, 7432, 7404, 7408,74192	2 each
2.	Power Supply	+ 5V	1
3.	LED with resistor	1kΩ	3 each

THEORY:

Counters: counters are logical device or registers capable of counting the no of states or no of clock pulse arriving at its clock input where clock is a timing parameter arriving at regular intervals of time, so counters can be also used to measure time & frequencies. They are made up of flip flops. Where the pulses are counted to be made of it goes up step by step & the o/p of counter in the flip flop is decoded to read the count to its starting step after counting *n* pulse incase of module & counters.

Synchronous Counter:
In this counter, all the flip flops receive the external clock pulse simultaneously.Ex:- Ring counter & Johnson counter
The gates propagation delay at reset time will not be present or we maysay will not occur.
Classification of synchronous counter:
Depending on the way in which counting processes, the synchronouscounter is classified is :-
1) *Up counter:*The up counter counts binary form 0 to7 i.e.(000 to 111).It counts from small to large number. It's O/P goes on increasing as they receive clock pulse.
2) *Down counter:*This down counter counts binary from 7-0 i.e.(111-000).It counts from large to small number. It's O/P goes on increasing as they receive clock pulse.
3) *Up down counter:*The up/down counter counts binary form 0 to7 *or* 7 to 0 depending on the Mode bitM = 0, it acts as Up counter and for M =1 as Down counter.

Procedure:

1. All the connections are made as per the circuit diagram.
2. Verify the output using the truth table.

Results:

The working of Mod N synchronous up and down counter verified using 7476, Mod-N counter using 7490 and 74192.

iii) Up/Down Counter

State Table for 3 bit Up-Down Synchronous Counter:

Control i/p M	Present State			Next State			Input for Flip-flop					
	Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	0	1	0	X	0	X	1	X
0	0	0	1	0	1	0	0	X	1	X	X	1
0	0	1	0	0	1	1	0	X	X	0	1	X
0	0	1	1	1	0	0	1	X	X	1	X	1
0	1	0	0	1	0	1	X	0	0	X	1	X
0	1	0	1	1	1	0	X	0	1	X	X	1
0	1	1	0	1	1	1	X	0	X	0	1	X
0	1	1	1	0	0	0	X	1	X	1	X	1
1	0	0	0	1	1	1	1	X	1	X	X	1
1	0	0	1	0	0	0	0	X	0	X	1	X
1	0	1	0	0	0	1	0	X	X	1	X	1
1	0	1	1	0	1	0	0	X	X	0	1	X
1	1	0	0	0	1	1	X	1	1	X	X	1
1	1	0	1	1	0	0	X	0	0	X	1	X
1	1	1	0	1	0	1	X	0	X	1	X	1
1	1	1	1	1	1	0	X	0	X	0	1	X

K-MAP

MQ_C \ Q_BQ_A	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$J_A = 1$

MQ_C \ Q_BQ_A	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$K_A = 1$

MQ_C \ Q_BQ_A	00	01	11	10
00	0	1	X	X
01	0	1	X	X
11	1	0	X	X
10	1	0	X	X

$J_B = M\overline{Q_A} + \bar{M}Q_A$

MQ_C \ Q_BQ_A	00	01	11	10
00	X	X	1	0
01	X	X	1	0
11	X	X	0	1
10	X	X	0	1

$K_B = M\overline{Q_A} + \bar{M}Q_A$

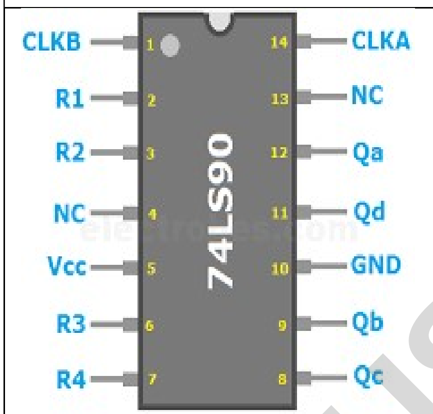
Q_BQ_A / MQ_C	00	01	11	10
00	0	0	1	
01	X	X	X	X
11	X	X	X	X
10	1	0		

Q_BQ_A / MQ_C	00	01	11	10
00	X	X	X	X
01			1	
11	1			
10	X	X	X	X

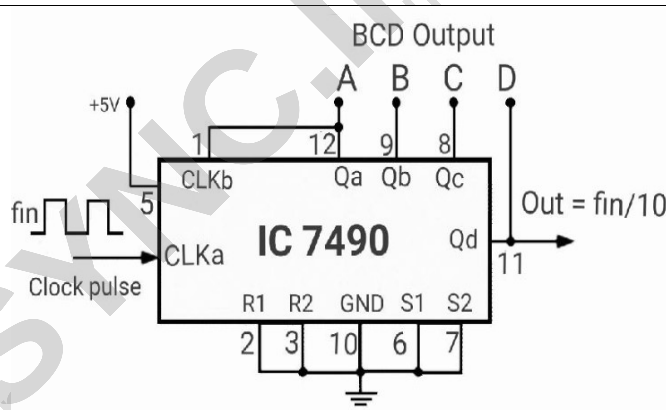
$$J_C = M\overline{Q_A Q_B} + \bar{M} Q_A Q_B \qquad K_C = M\overline{Q_A Q_B} + \bar{M} Q_A Q_B$$

b) Mod-N Counter using IC7490 / 7476.

7490 Pin Diagram

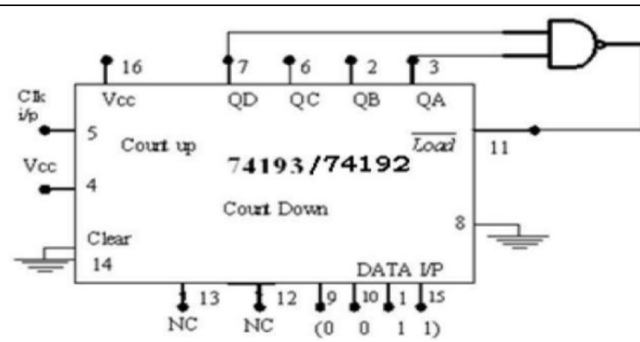


Mod-10 Counter



c) Synchronous counter using IC74192.

CLK	Q_D	Q_C	Q_B	Q_A
0	0	0	1	1
1	0	1	0	0
2	0	1	0	1
3	0	1	1	0
4	0	1	1	1
5	1	0	0	0
6	0	0	1	1



Pre Lab Questions::

1.What do you mean by Counter?

2.What are the types of Counters? Explain each

3.What are the applications of synchronous counters?

Assignment Questions:

1.What are the advantages of synchronous counters over asynchronous counters?

2. Ring counter is an example of synchronous counters or asynchronous counter?

3.Twisted Ring (Johnson's) counter is an example of synchronous counters or asynchronous counter?

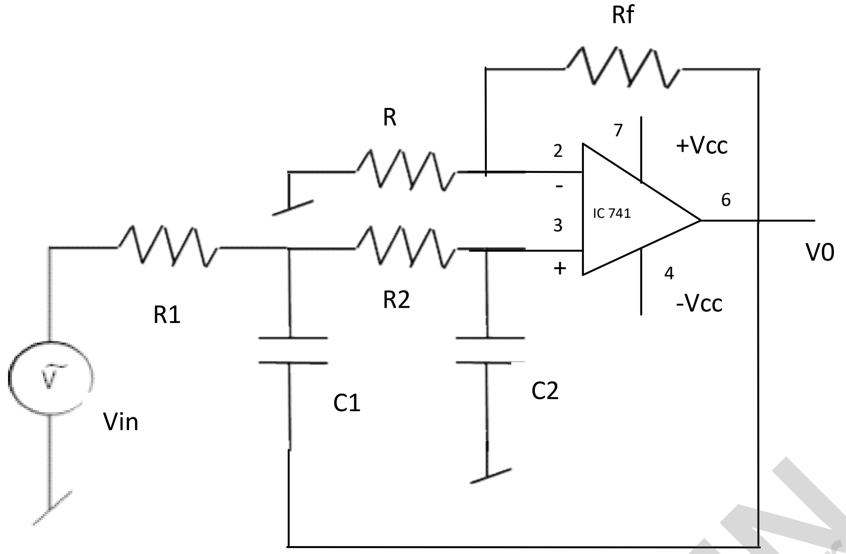
4. What is the difference between ring counter and twisted ring counter?

DEMONSTRATION EXPERIMENTS

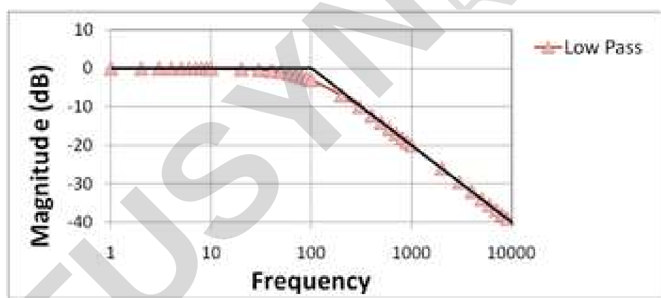
EXPERIMENT: 9

ACTIVE FILTERS

CIRCUIT DIAGRAM:



NATURE OF GRAPH:



AIM: Design II-Order Butterworth Low pass filter for a given cut off frequency **f_c=1kHz**, conduct the experiment by verifying frequency response.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1v to 80v and 1Hz to 260GHz	1
2.	Signal Generator	0.1v to 40v and 1Hz to 1GHz	1
3.	Power Supply	+/- 12v	1
4.	Op-Amp	741 +/- Vsat = 15v	1
5.	Resistor	1K, 10K	4,1
6.	Capacitor	0.01uF	2

THEORY:

The Low pass filter is one which allows the lower frequency signal and blocks the higher frequency signals. The signals which are allowed in the low pass filter are under pass band region, the signals which are blocked in filter are under stop band region. In pass band the signal strength will be high and after the cut off frequency signal strength starts decreasing which is called roll off of the signal strength.

The order of the filter will increase the roll off between the Pass band and stop band region.

PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Apply the sine wave input of peak voltage = ____v.
3. Keep the input voltage constant and vary the frequency until the output voltage start reducing (0.707 Vo).
4. Note down the reading of input and output voltage levels starting from 100 Hz till the designed cut off frequency and, take some more reading after the cut off frequency to show the clear roll off the low pass filter after cut off frequency.

DESIGN:

Assume the Gain A_v = 2 , cut off frequency f_c is given

1. $A_v = 1 + (Rf/R) = 2$ then Rf = R, choose Rf = R = 10kΩ

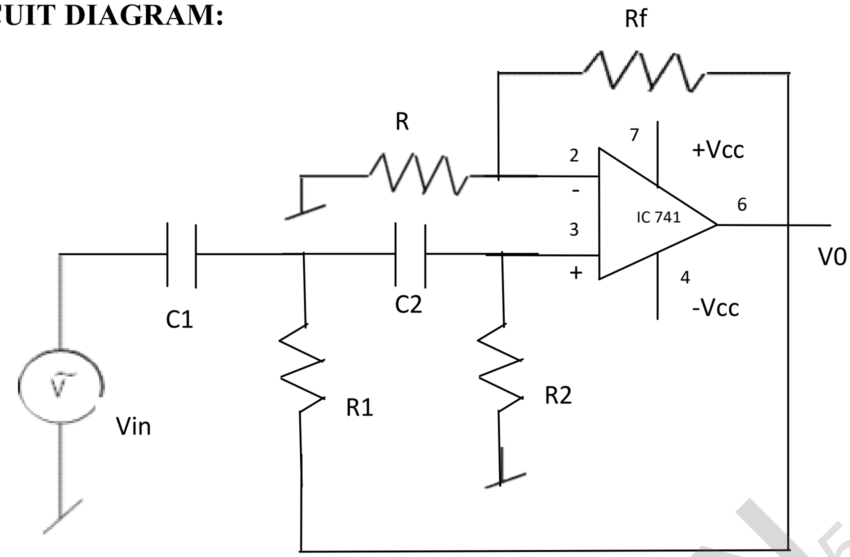
2. Filter: $f_c = 1/[2\pi(R_1C_1R_2C_2)^{1/2}]$ let C1= C2 = C and R1 = R2 = R then f_c equation simplified to $f_c = 1/[2\pi RC]$. Assume C = 0.22 µF and

   $R = [1/2\pi f_c C]\ \Omega$.

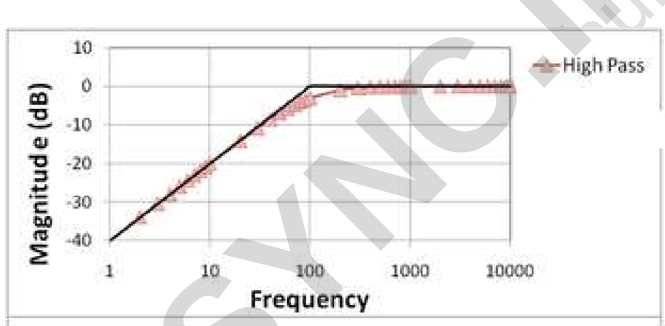
   $R = [1/2\pi(10^3)(0.22 \times 10^{-6})]\ \Omega$

   R = <u>723.432</u> Ω ≈ <u>750</u> Ω

CIRCUIT DIAGRAM:



NATURE OF GRAPH:



THEORY:

The High pass filter is one which blocks the lower frequency signal and allows the higher frequency signals. The signals which are allowed in the High pass filter are under pass band region, the signals which are blocked in filter are under stop band region. In pass band the signal strength will be low and after the cut off frequency signal strength starts Increasing.

The order of the filter will increase the roll off between the Pass band and stop band region.

PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Apply the sine wave input of peak voltage = ____v.
3. Keep the input voltage constant and vary the frequency until the output voltage start increasing (0.707 Vo).
4. Note down the reading of input and output voltage levels starting from 100 Hz till the designed cut off frequency and, take some more reading after the cut off frequency to show the clear roll off the High pass filter after cut off frequency.

DESIGN:

Assume the Gain $A_v = 2$, cut off frequency f_c is given

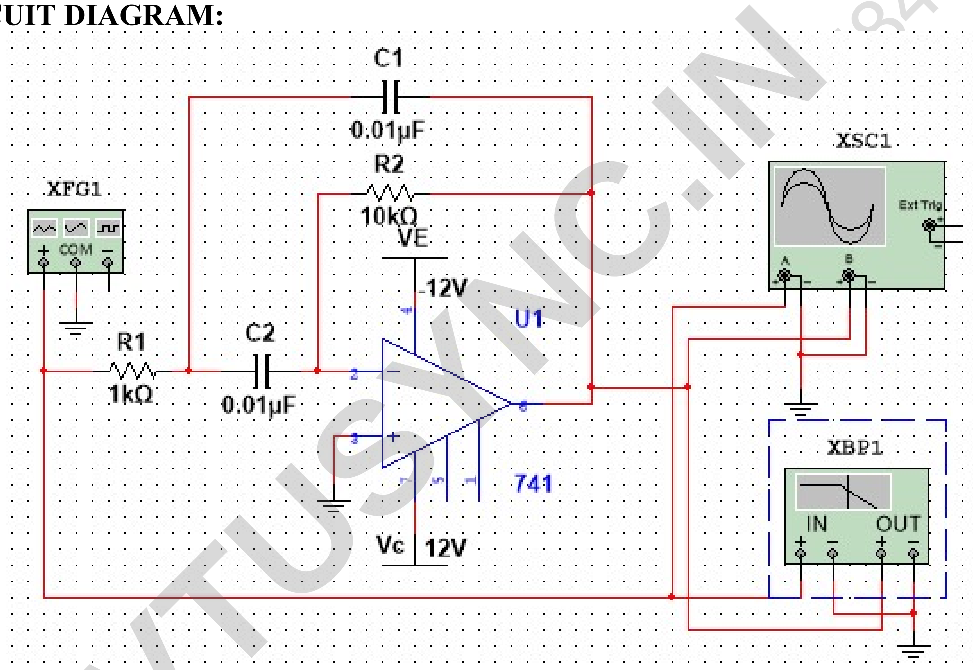
1. $A_v = 1 + (Rf/R) = 2$ then $Rf = R$, choose $Rf = R = 10k\Omega$
2. Filter: $f_c = 1/[2\pi(R_1C_1R_2C_2)^{1/2}]$ let C1= C2 = C and R1 = R2 = R then f_c equation simplified to $f_c = 1/[2\pi RC]$. Assume C = 0.01 µF and

   $R = [1/2\pi f_c C]\ \Omega$.

   $R = [1/2\pi(5x10^3)(0.47x10^{-6})]\ \Omega$

   R = <u>67.72</u> Ω

CIRCUIT DIAGRAM:

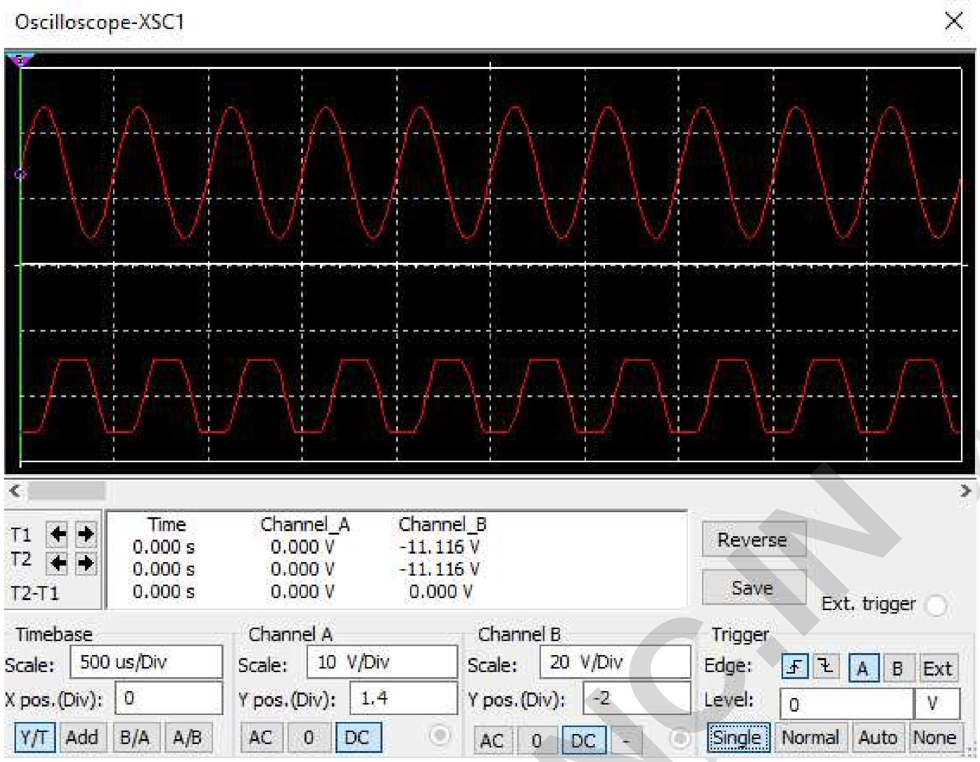


Design: $R_1 = 3.3\ k\Omega$, $R_2 = 100\ k\Omega$, C = 0.01 µF

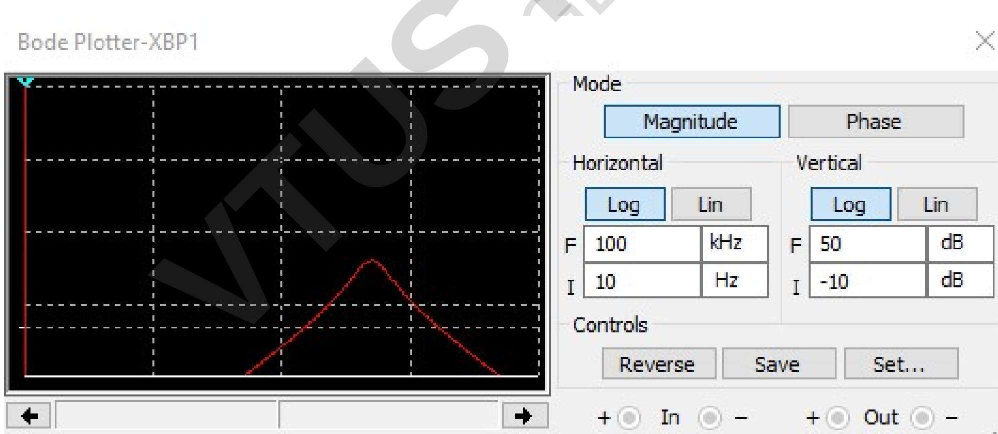
$$f1 = \frac{1}{2\pi\ RC} = \frac{1}{2\pi\ 1k\ 0.01\mu} = kHz$$

$$f2 = \frac{1}{2\pi\ RC} = \frac{1}{2\pi\ 10k\ 0.01\mu} = kHz$$

Output:



Output: Magnitude Plot



Theory:

The narrow Band Pass Filter Circuit using multiple feedback,as shown in this circuit, the filter uses only one opamp. This filter is unique in the following respects.It has two feedback paths, which is why it is called a multiple feedbackThe opamp is used in the inverting mode. Generally a narrow band pass filter is designed for specific values of centre frequency fc and Q, or fc and band width.

Procedure:

i. All the connections are made as per the circuit diagram.

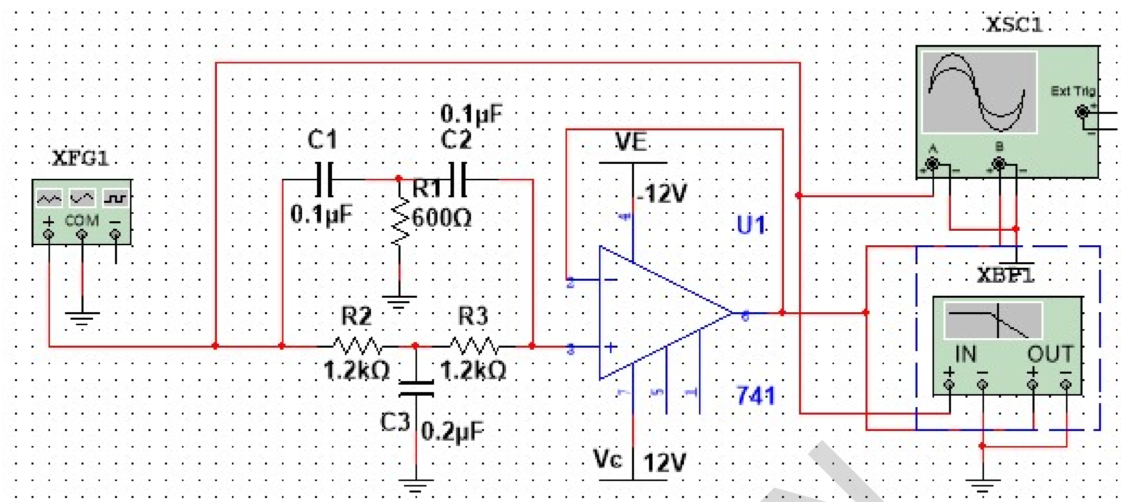
ii. Resistor and Capacitor values used as per the Designed values.

iii. Adjust the Oscilloscope to observe the output.

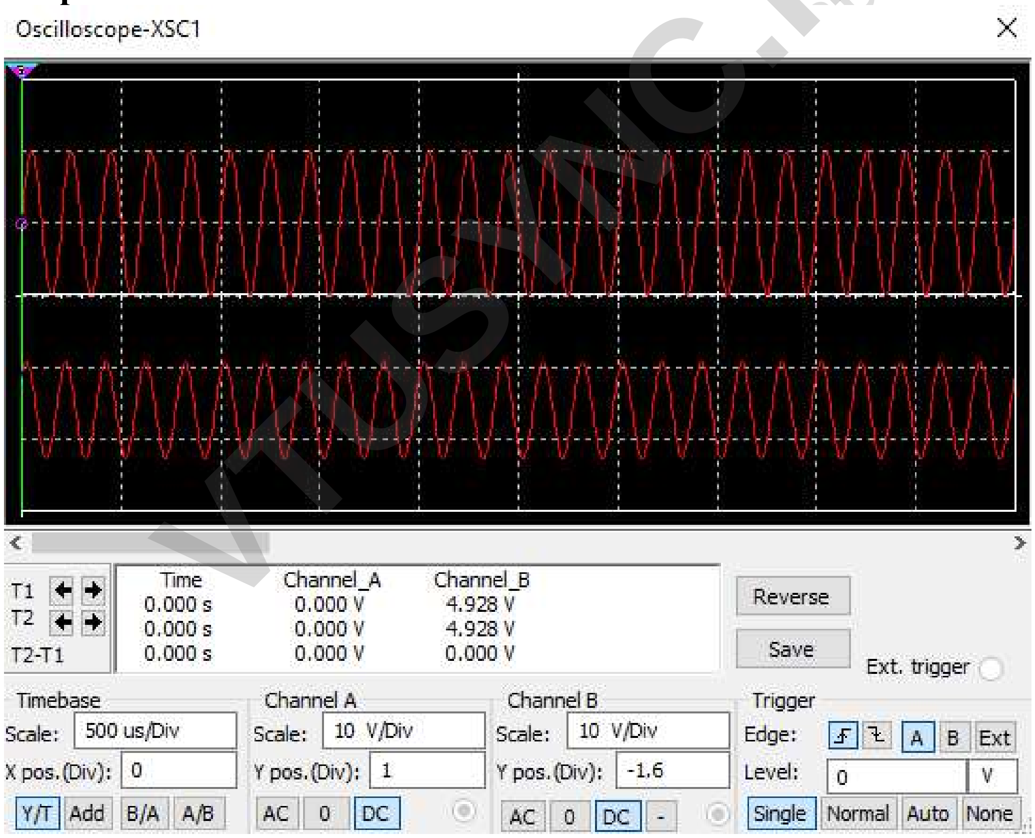
iv. Use Bode plotter to observe the Frequency plot.

v. Compare the results of Bode plotter with the cut off frequency of the Filter.

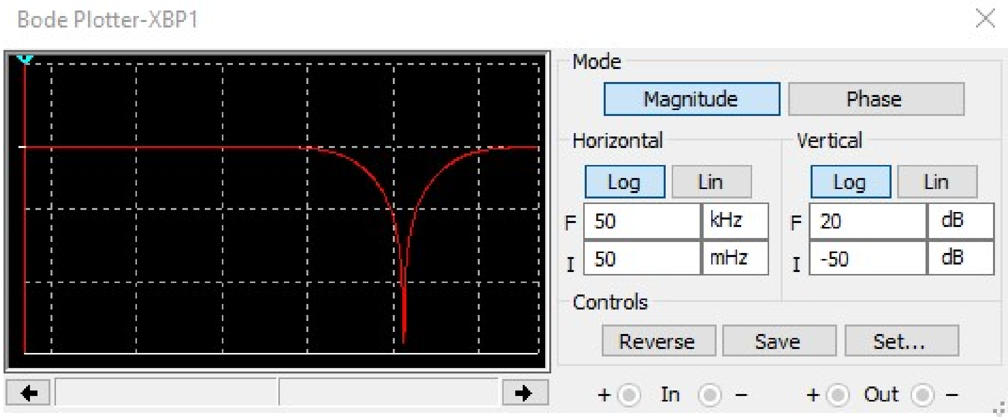
CIRCUIT DIAGRAM:



Output:



Output: Magnitude Plot



Theory:

The narrow band reject filter, often called the notch filter, is commonly used for the attenuation of a single frequency. For example, it may be necessary to attenuate 60 Hz or 400 Hz noise or hum signals in a circuit. The most com-monly used notch filter is the Twin T network. One T network is made up of two resistors and a capacitor, while the other is made of two capacitors and a resistor. The frequency at which maximum attenuation occurs is called the notch-out frequency. One disadvantage of the passive twin T network is that it has a relatively low figure of merit, Q. As discussed earlier, the higher the value of Q, the more selective is the filter. Therefore, to increase the Q of the twin T network sig-nificantly, it should be used with a voltage follower. The Notch filters are used in communications, biomedical instruments, etc. where the elimination of certain frequencies is necessary.

Procedure:

i. All the connections are made as per the circuit diagram.

ii. Resistor and Capacitor values used as per the Designed values.

iii. Adjust the Oscilloscope to observe the output.

iv. Use Bode plotter to observe the Frequency plot.

v. Compare the results of Bode plotter with the cut off frequency of the Filter.

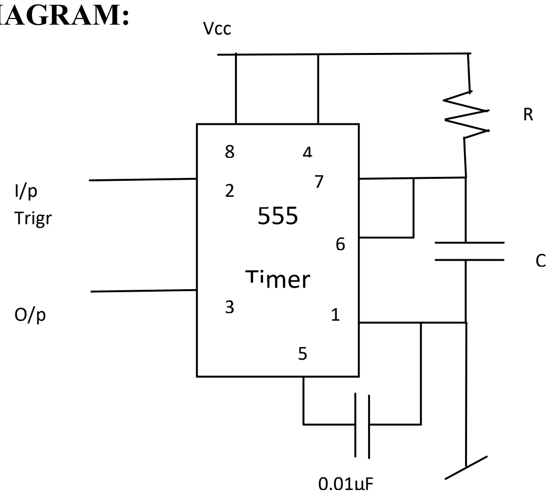
RESULTS:

The Active Filter Circuits Simulation results verified with Designed value:

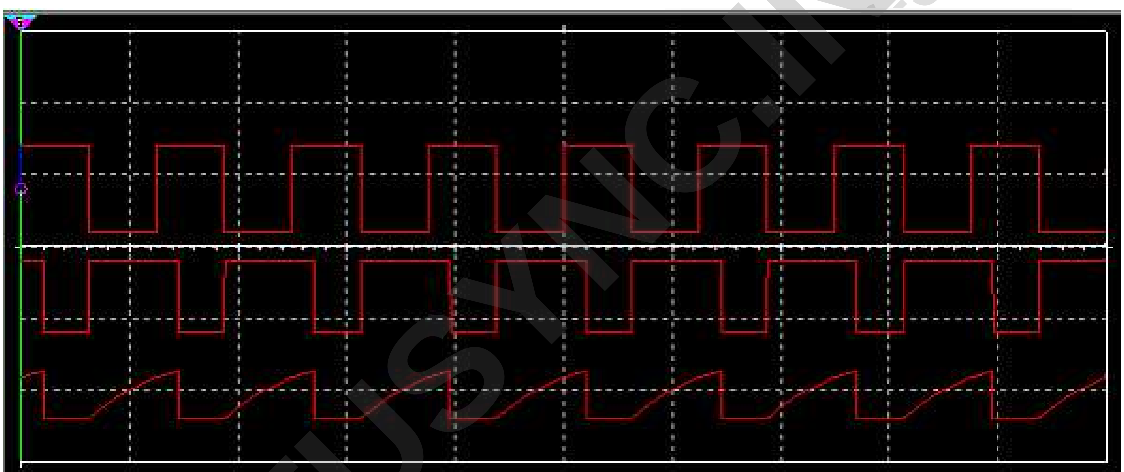
EXPERIMENT: 10

Multivibrator using 555 timer (Monostable & Astable)

CIRCUIT DIAGRAM:



NATURE OF WAVEFORMS:



OBSERVATION:

Output Time period of the pulse = ______ msec

Output voltage of the pulse = ______ V

AIM: Using 555 timer Design Monostable multivibrator for a given pulse width time of **1msec**.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1v to 80v and 1Hz to 260GHz	1
2.	Signal Generator	0.1v to 40v and 1Hz to 1GHz	1
3.	Power Supply	5v	1
4.	TimerIC	555	1
5.	Resistor	2 kΩ	1
6.	Capacitor	0.47, 0.01 μF	1, 1

THEORY:

A monostable multivibrator is a pulse generating circuit in which the duration of the pulse is determined by the RC network connected external to 555 timer. When external trigger pulse is applied, the output is forced go high. The time output remain high is determined by the external RC network. At the end of timing interval, the output automatically reverts back to low state. It remains low until the trigger pulse is applied again, then the cycle repeats.

The monostable circuit has only one stable state, hence it is called monostable. Generally the output of monostable circuit is LOW.

PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Adjust the input trigger source and observe the output from the timer.

DESIGN:

Given the pulse width T = 1 msec

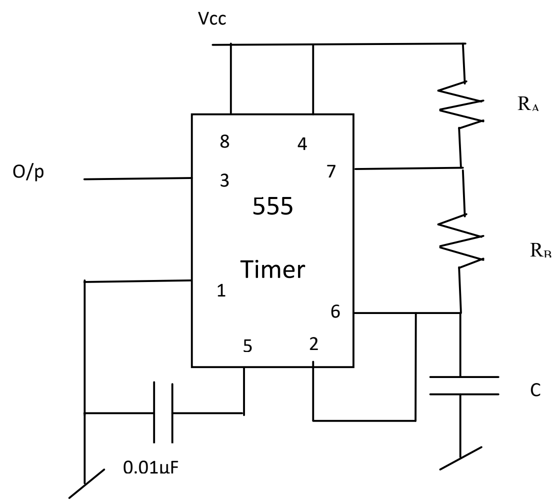
$T = 1.1\ RC$

Assume C = 0.47 μF

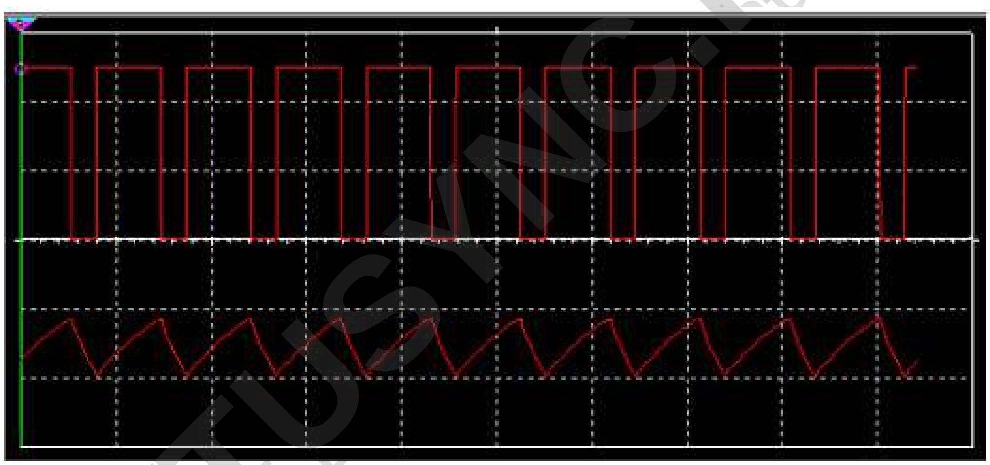
$R = T/(1.1C)$

$R = \underline{1934.23}\ \Omega \approx 2\ k\Omega$

CIRCUIT DIAGRAM:



NATURE OF WAVEFORMS:



OBSERVATION:

Output Voltage across Capacitor = ______V

Total Output voltage (at pin no. 3) = ______V

T_{on} =______msec

T_{off} = ______msec

AIM: Using 555 timer Design Astable multivibrator for a given duty cycle 70% for a frequency of 1 kHz.

COMPONENTS & APPARATUS:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	CRO	0.1v to 80v and 1Hz to 260GHz	1
2.	Signal Generator	0.1v to 40v and 1Hz to 1GHz	1
3.	Power Supply	5v	1
4.	Timer IC	555	1
5.	Resistor	5.79, 4.34, kΩ	1,1
6.	Capacitor	0.01, 0.1 μF,	1,1

THEORY:

An Astable multivibrator is called free-running multivibrator, in this circuit external trigger to change the output state is not required. The time interval for ON and OFF is determined by the two resistor's and capacitor.When the output is high the capacitor C starts charging towards Vcc through RA and RB. As soon as capacitor equals (2/3)Vcc, comparator 1 triggers flip-flop , and output switches low. Capacitor C starts discharging through RB and transistor Q1. When voltage across C equals (1/3)Vcc, comparator 2's output triggers and output goes high.

PROCEDURE:

1. All the connections are made as per the circuit diagram.
2. Adjust the Resistor R_A and R_B and observe the output waveforms of the timer.

DESIGN:

$T_{on} = 0.693(R_A+R_B)C$ $\qquad T_{off} = 0.693\,(R_B)C$

$T = T_{on} + T_{off}$

Given that $f_c = 1\text{kHz} = [1.45/(R_A + 2R_B)C]$

%duty cycle $= (T_{on}/T)\,100 = 70\%$ $\qquad T = 1/f_c = 1\text{msec}$

$= [(R_A+R_B)/(R_A+2R_B)]\,100$

$T_{on} = (0.70)1\text{msec} = 0.7$ msec

$T_{off} = 1\text{msec} – 0.7\text{msec} = 0.3$ msec

Let assume C = 0.1 μF

$R_B = (0.3\text{x}10^{-3})/(0.69\text{x}0.1\text{x}10^{-6}) = \underline{4.3478}\ \text{k}\Omega$

$R_A = (0.7\text{x}10^{-3})/(0.69\text{x}0.1\text{x}10^{-6}) - 4.3478\ \text{k}\Omega = \underline{5.79}\ \text{k}\Omega$

RESULT:

Monostable multivibrator is designed for the pulse width T = ______ msec and practical observed value of pulse width T = ____ msec.

Astable multivibrator is designed for frequency f = ______kHz and duty cycle T_{on} = _____msec and T_{off} = ________ msec, and observed frequency is f = ______kHz and duty cycle T_{on} = ________ msec and T_{off} = _______msec.

Pre Lab Questions:

1. What is Astable & Monostable Multivibrator?

2. What is comparator?

Lab Assignment Questions:

1. What are applications of Multivibrator?

2. What is the action of Capacitor in Multivibrator?

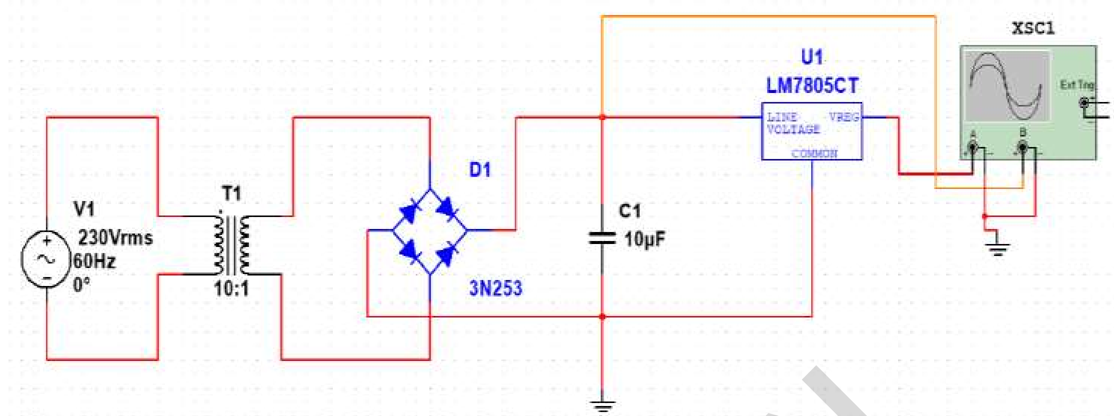
3. Explain the comparator operation?

4. what is necessity of bypass capacitor at pin 5?

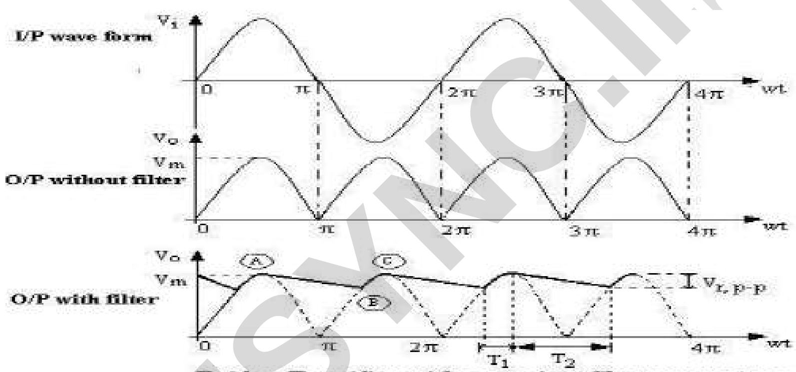
EXPERIMENT: 11

Regulated Power Supply

CIRCUIT DIAGRAM:



Waveforms:



Bridge Rectifier with capacitor filter wave form

Simulator Output:

AIM: Design Regulated power supply

COMPONENTS & APPARATUS in Simulator:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	Diodes	IN4007	4
2.	StepDown Transformer	230V/ 0-12V	1
3.	3 Pin fixed voltage regulator	IC7805	1
4.	Resistor/ Pot	100Ω/ 1KΩ	1,1
5.	Capacitor	470 µF	1

THEORY:

A regulated power supply converts unregulated AC (Alternating Current) to a constant DC (Direct Current). A regulated power supply is used to ensure that the output remains constant even if the input changes. The input is given to the stepdown transformer. A step down transformer will step down the voltage from the ac mains to the required voltage level. The output of the transformer is given as an input to the rectifier circuit. Rectifier is an electronic circuit consisting of diodes which carries out the rectification process. Rectification is the process of converting an alternating voltage or current into corresponding direct (DC) quantity. The input to a rectifier is AC whereas its output is unidirectional pulsating DC. The rectified voltage from the rectifier is a pulsating DC voltage having very high ripple content. Hence a capacitor filter is used to remove the ripple in the rectified output . The filtered output is given to the regulator circuit. A regulator will maintain the output constant even when changes at the input or any other changes occur. IC's like 78XX and 79XX (such as the IC 7805) are used to obtained fixed values of voltages at the output.

PROCEDURE:

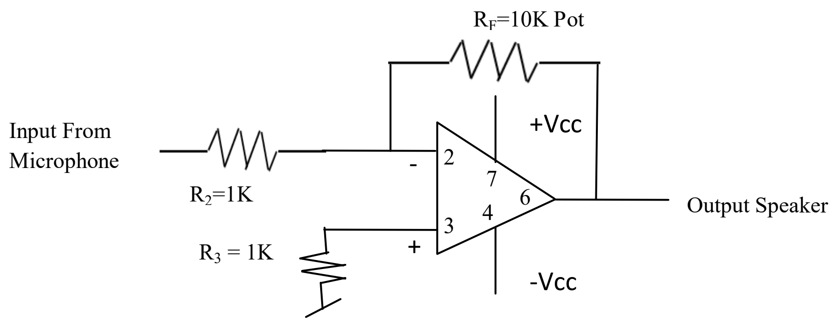
1. Construct the circuit in multisim as shown in circuit diagram.
2. Click on the RUN button and use the CRO to observe the waveform.

RESULT: The Regulated power supply verified.

EXPERIMENT: 12

Audio Amplifier

CIRCUIT DIAGRAM:



AIM: Design and test an audio amplifier by connecting a microphone input and observe the output using a loud speaker.

COMPONENTS & APPARATUS in Simulator:

Sl. No.	Component and Apparatus	Specification	Quantity
1.	Op-Amp	741	1
2.	Power Supply	+/- 12V	1
3.	Resistor/ Pot	10KΩ / 10KΩ	1,1
4.	Capacitor	10 μF	1
5.	Microphone, Speaker	8 Ω	1 each

PROCEDURE:

Audio refers to the representation of sound which can be perceived by humans. The electronics can be used to receive audio signals (via microphone), record audio in some storage, transmit audio (through wired or wireless communication channels) and reproduce audio signals (via speakers). The audio can be represented and transmitted as either analog signals or digital signals. In this series, analog audio signals are the concern. The audio signals have a frequency range of 20 Hz to 20,000 Hz.

If an electrical signal carrying audio is transmitted from one end and received at another end one mile away, it loses 90 percent of its strength. losses whether the signal is transmitted from just the microphone to the recording device, computer or audio generator to a speaker or it is transmitted on wire over a long distance. In order to sort out this problem, engineers devised special electronics – 'Amplifiers'. An Amplifier increases the strength of an audio signal by increasing its amplitude. The rise in amplitude is called Amplification. That is why, it is called Amplifier. An audio amplifier needs to be designed based on its application and required specifications.

RESULTS:

Audio Amplifier designed and tested.